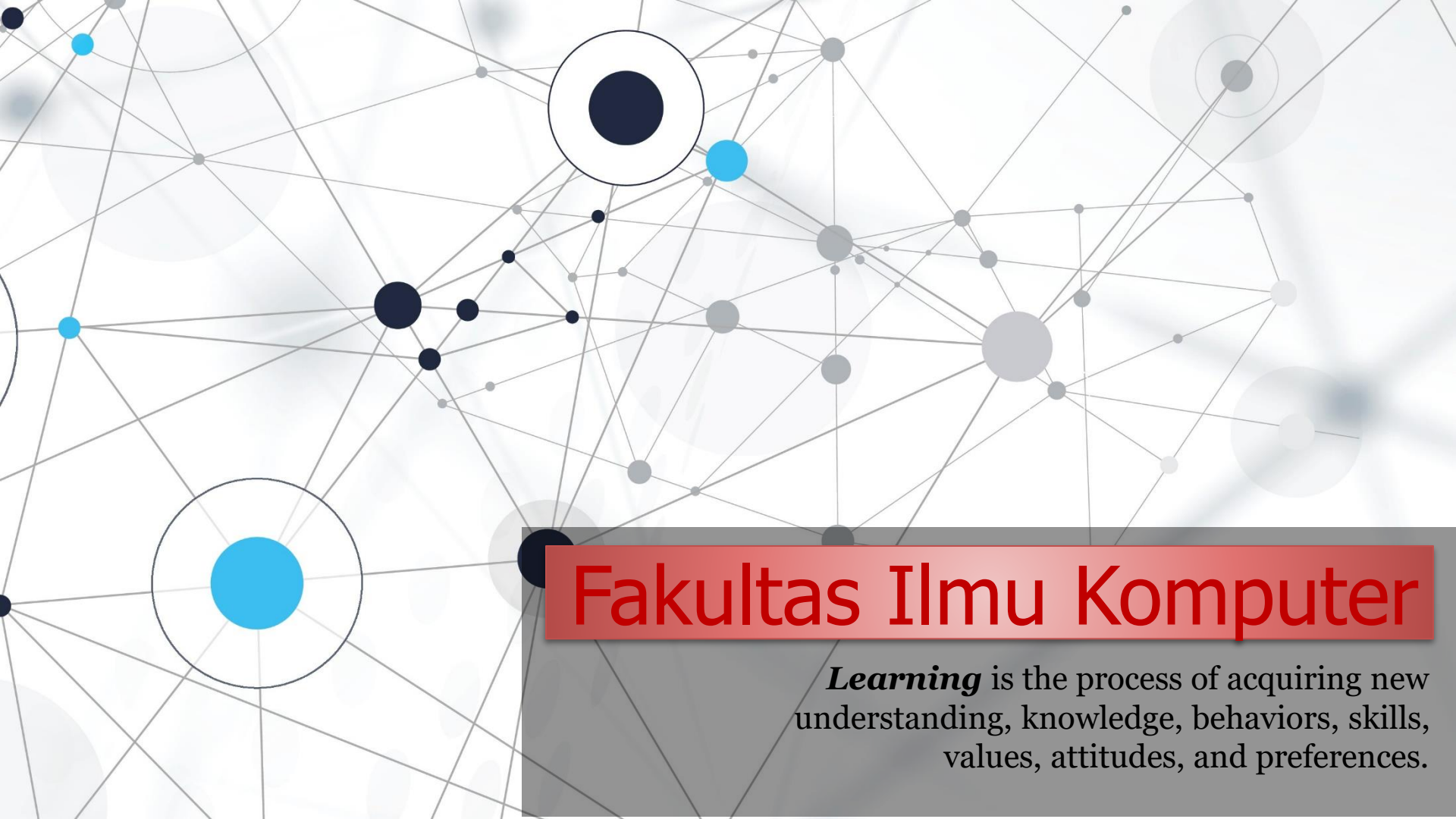


An abstract network diagram with various sized nodes (black, blue, grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

**Server Side Programming Languages
(PHP, Python, Ruby, *etc.*)**



Welcome to *the*

DevOps Engineering *Course*

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1 - Back-end Web Development

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Jenis-Jenis Web Development

FRONT-END WEB DEVELOPMENT

Tugas utama:

- ❖ Para developer pada bagian ini bertugas untuk membangun UI yang dapat membantu pengguna (*users*) untuk mencapai tujuan mereka di dalam situs.
- ❖ **Front-end web developer** juga sering ikut dalam aspek perancangan User Experience (UX) dalam project.
- ❖ Memiliki *background* di bidang UX dapat menjadi nilai plus bagi seorang front-end developer.



BACK-END WEB DEVELOPMENT

Tugas utama:

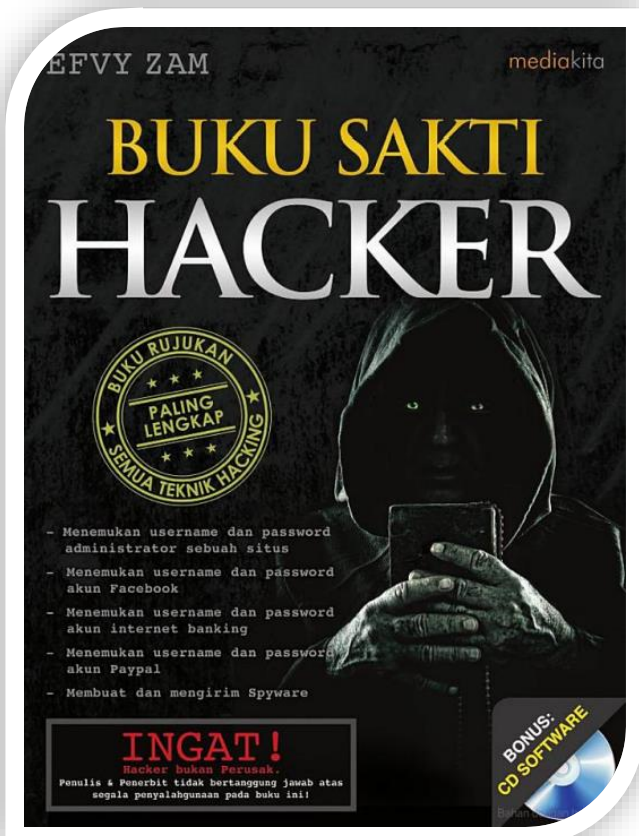
- ❖ Back-end developer bertugas untuk **mengelola server situs web**, program (*database*), dan software, dengan tujuan agar semua fitur tersebut dapat bekerja dengan baik.
- ❖ **Back-end developer** bekerja dalam sistem seperti ***server***, *sistem operasi*, **API**, dan *basis data*.
- ❖ Bertugas untuk mengelola *code* dalam situs untuk **keperluan keamanan**, *konten*, dan *struktur situs*.



Apakah ada sistem yang benar-benar aman 100%?

Port Scanning

4

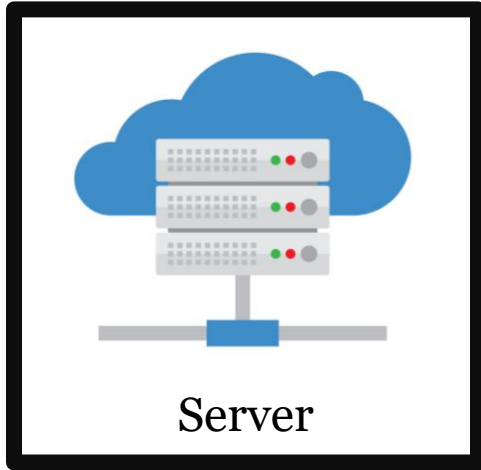


Jika diibaratkan sebuah rumah, port adalah pintu dan jendela rumah tempat keluar masuknya data. Secara logika, tidak ada sistem yang aman 100%. Apabila sebuah sistem aman 100%, tentu saja semua pintu dan jendela akan ditutup semua. Ibaratnya, jika pintu internet tidak dibuka, Anda pun tidak akan bisa menghubungkan komputer dengan internet. Jadi, bisa kita katakan bahwa port adalah pintu keluar masuknya paket data. }

Secara garis besar, port dapat dibagi dua bagian. Yang pertama adalah port fisik (*physical port*) yang merupakan port di bagian belakang CPU, seperti port serial, dan port monitor. Lalu ada pula port perangkat lunak (*software port*), merupakan port yang digunakan oleh software untuk melakukan koneksi dengan komputer lain.

Port juga mengidentifikasi sebuah proses tertentu dimana sebuah server dapat memberikan sebuah layanan kepada klien atau bagaimana sebuah klien dapat mengakses sebuah layanan yang ada dalam server. Ada banyak port yang terdapat pada sebuah komputer, apalagi sewaktu terhubung dengan internet. Beberapa port yang umum adalah port 80 (HTTP), yaitu port untuk membuka sebuah halaman website, port 20 (FTP) untuk melakukan upload maupun download file, port 110 (POP3) untuk menerima email. Serta masih banyak jenis port lainnya.

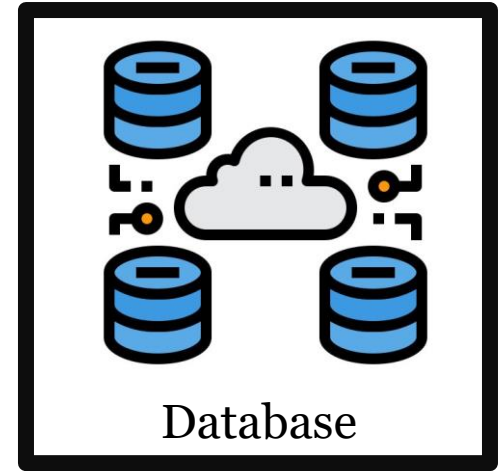
Skills & Tools Required *for* Back-End



Server

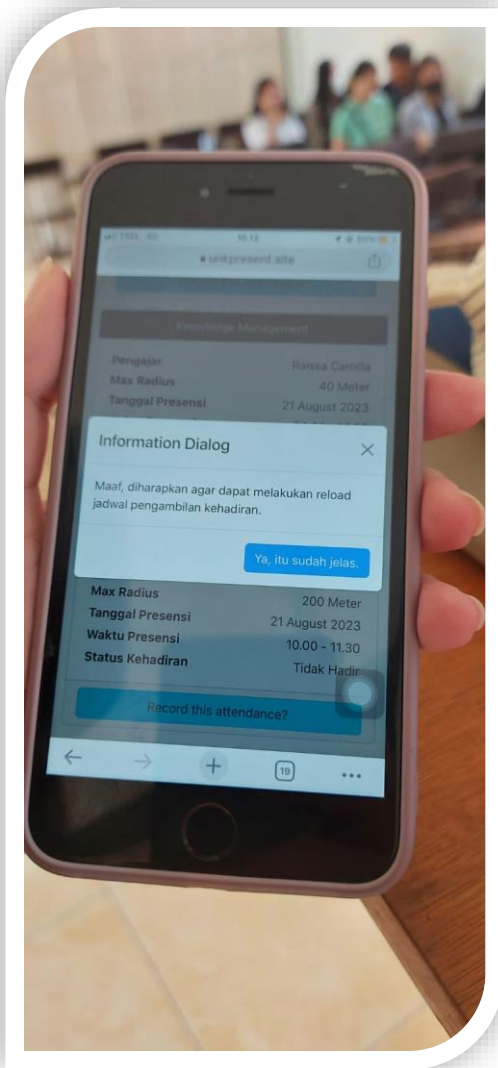


Application



Database

In order to make the server, application and database communicate with each other, back-end developers use server-side languages like **PHP**, Ruby, **Python**, Java and .Net to build an application and tools like MySQL, Oracle and SQL Server to find, save or change data and serve it back to the *user* in front-end code.



Untuk mencari
error atau *bug*
hampir sama
seperti **mencari**
seekor kutu.



Jika itu tentang pengujian performa sistem, maka *stress test* perlu dilakukan.

Beban Kerja yang Berat

Kondisi Ekstrem

Tekanan Tinggi

Potensi Kelemahan

Monitoring & Log Management

Seorang ***back-end web developer*** sebaiknya mengetahui tentang *auth log*, *error log*, *access log*, dan *slow log* sebagai catatan atau rekaman aktivitas di sebuah server, terutama jika mereka bekerja dengan server dan aplikasi web.

Auth Log (Authentication)

Mencatat potensi pelanggaran keamanan di dalam lingkungan server.

Error Log (Debugging)

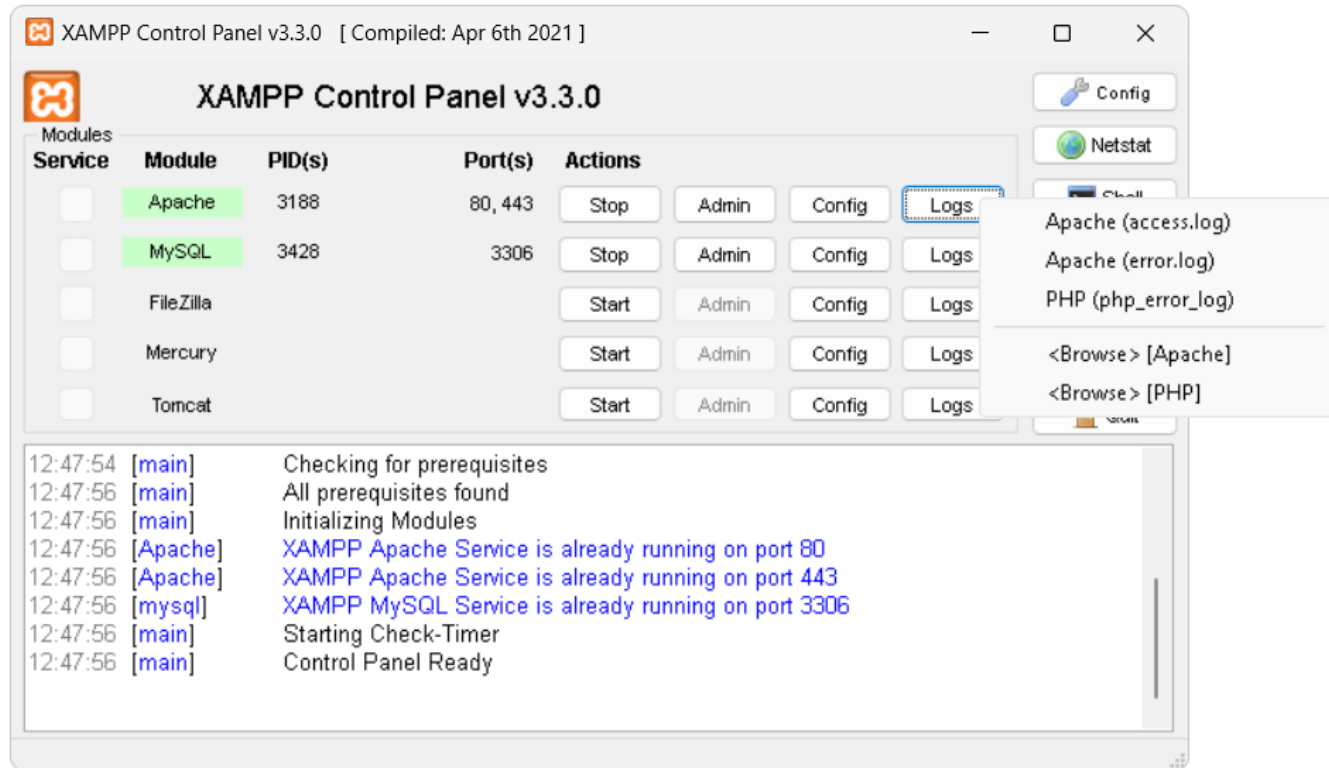
Error log mencatat/mendiagnosis masalah yang terjadi pada sistem

Access Log (Monitoring)

Access log mencatat semua permintaan yang diterima oleh server.

Slow Log (Performance)

Slow log mencatat/mengidentifikasi query yang lambat di database (MySQL).

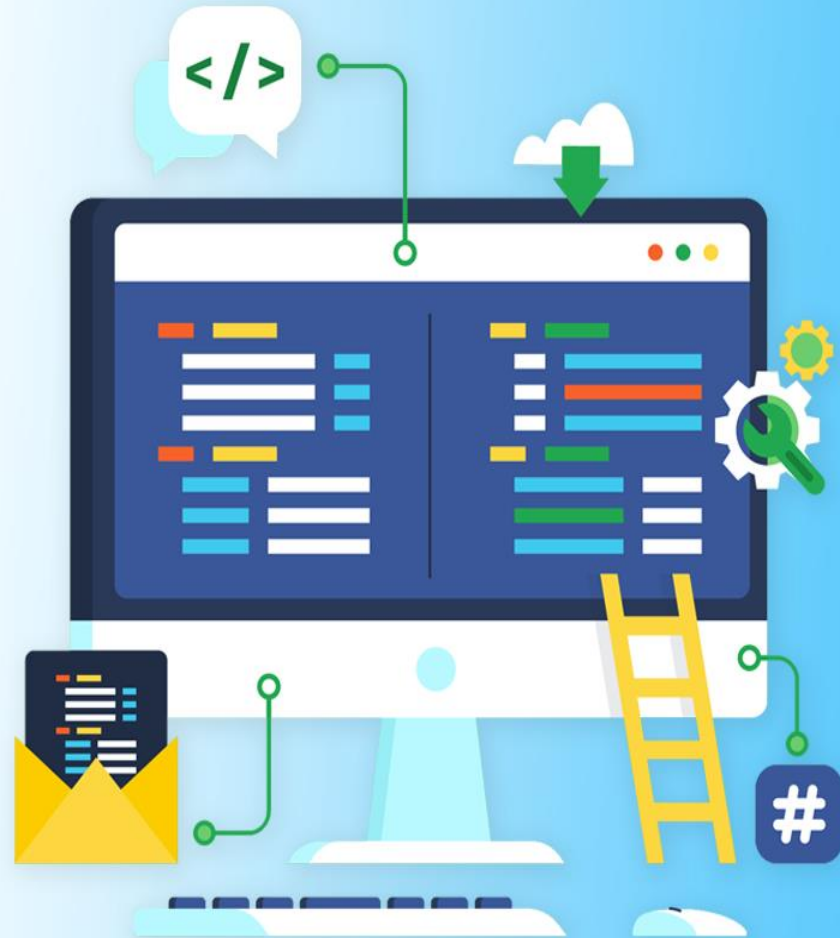
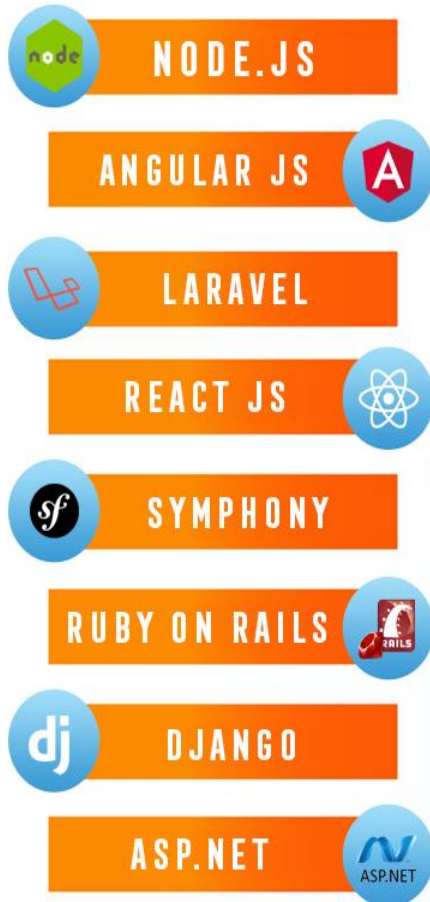


In XAMPP Apache on Windows, the **error.log** and **access.log** files are used to record server activities and provide important information related to Apache operations. The **error.log** file logs all error messages or warnings generated by the Apache server, while the **access.log** file records all HTTP requests received by the Apache server.

Main Uses:

- 1) Track the source of issues or bugs on the server.
- 2) Analyze the causes of server process failures.
- 3) Analyze user access patterns on the server.
- 4) Monitoring server activity to detect potential abuse (**brute-force attacks** or **unauthorized access**).
- 5) Identify the most frequently accessed content.

TOP **WEB** DEVELOPMENT FRAMEWORKS



Framework MVC

Bahasa	Framework
PHP	CodeIgniter, Laravel, Yii, dll
Java	Spring MVC, JSF, Struts, dll
Python	Django, CherryPy, dll
Ruby	Ruby on Rails, Sinatra ,dll
Javascript	AngularJS, React, Backbone.js, dll

Seorang DevOps Engineer harus terbiasa dengan banyak jenis programming languages, tools dan juga frameworks.

Bagaimana cara agar bisa tahu dan mengerti banyak bahasa pemrograman dan juga frameworks? Apakah ini membutuhkan banyak waktu juga di pelajari satu per satu?



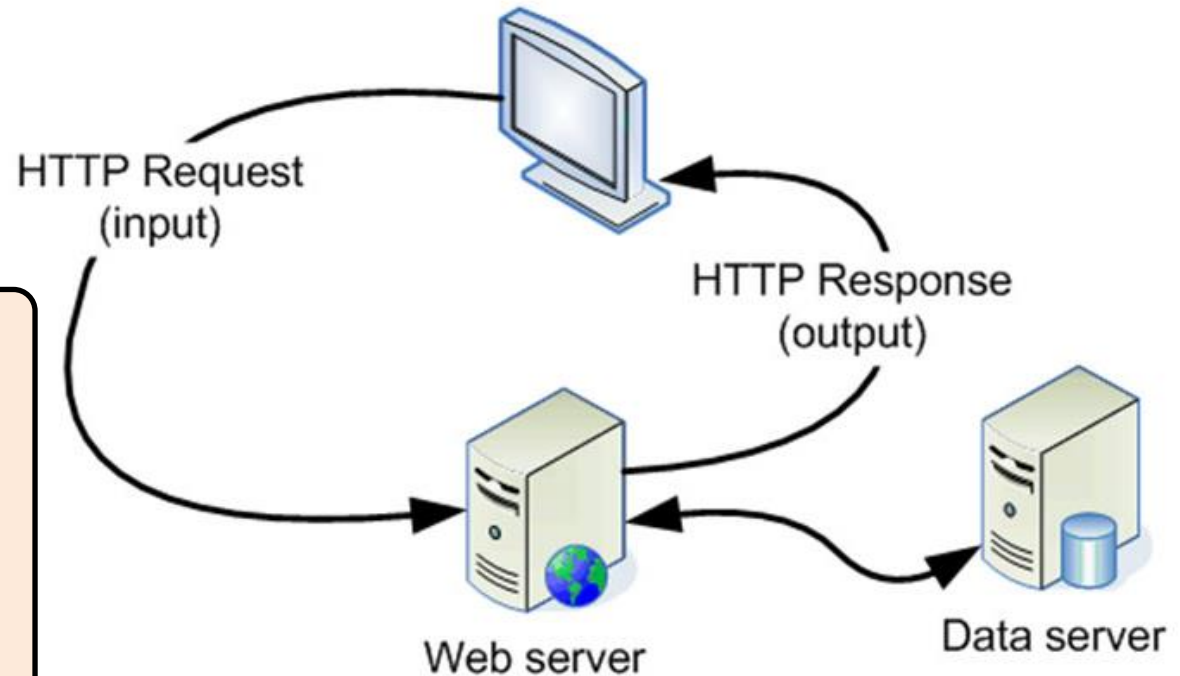
The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

2 - Server-side Scripting Languages

What is Server-side Scripting?

If we want to :

- 1) We want to view a webpage.
- 2) The browser sends a request to the webserver.
- 3) The server validates and processes the request.
- 4) The server sends the content of the page back.



- ❖ Server-side scripting is a programming language that runs on the webserver.
- ❖ Server-side scripting is also called “*backend*” programming because it happens behind the scenes on the backend of the website.

Server-side Script vs Client-side Script

Client-side scripting (*front-end*) is executed directly in the browser.

Frameworks like Angular, Vue, and React fall into this category.

The client script will communicate with the **server through APIs**, but the user interface processing is done in the browser.

Server-side scripting is a programming language that runs on the **web server**.

Server-side scripting is also called “*back-end*” programming because it happens behind the scenes on the backend of the website.

Server-side Programming



- 1) Querying the database
- 2) Operations over databases
- 3) Access/Write a file on server.
- 4) Interact with other servers.
- 5) Structure web applications.
- 6) Process user input.

Benefits Server-side Scripting

Scalability

Pengembangan web dapat menjadi sangat menantang jika kita memiliki banyak sekali *users* yang mengunjungi website kita; semakin banyak *users* yang berkunjung, semakin banyak sumber daya (*resource*) yang dibutuhkan oleh aplikasi web.

Bahasa pemrograman sisi server akan memastikan bahwa web server dapat menangani permintaan traffic secara efisien.

Hal ini akan memungkinkan kita untuk menangani data traffic yang sangat besar tanpa perlu khawatir situs web akan lemot (*website crashing*).

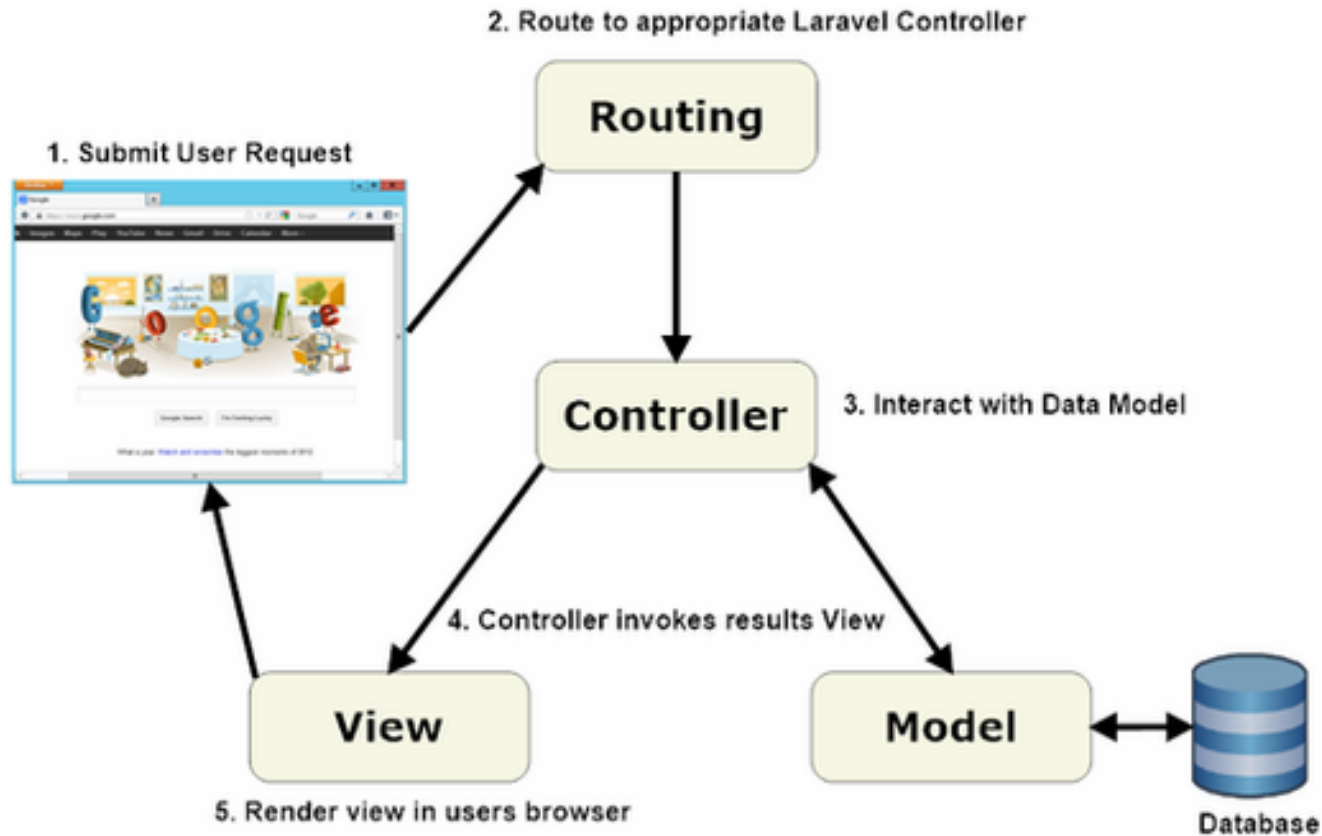
Benefits Server-side Scripting

Security

Server-side scripting language adalah bagian penting dari manajemen keamanan aplikasi web karena menyediakan validasi input dan **output encoding, mencegah sebagian besar serangan injeksi kode.**

Server-side scripts juga berguna untuk melacak profil pengguna, menghapus spam, dan **bahkan memblokir lalu lintas berdasarkan alamat IP.**

Laravel Controller



MVC URL Routing system allows to create powerful and flexible URLs for the applications.

It provides two functionalities:

- 1) **Inspect incoming URLs** - This functionality manifests when the application receives a client request.
- 2) **Generate outgoing URLs** - This functionality manifests when we render URLs using Url Action in the View.

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3 - Server-side Scripting Languages for Web Development

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Server-side scripting languages are a type of programming language that **runs on the server** instead of the client. These programming languages can create dynamic websites and web applications, perform scheduling and data mining tasks, automate processes such as compilation, and send emails.

PHP



The **PHP interpreter** is found on more than 20 million websites. The flexibility of PHP makes it easy to integrate with many different databases, including SQL and MySQL, and independent of any particular database system.

In addition to its flexibility, PHP is notable for being open-source software. Most web servers use PHP to implement the Basic and Configuration Directives (known as Apache mod_php). Because of its popularity, the demand for PHP programmers is high. PHP also has several high-profile users, including Facebook and Wikipedia.

Node.js



Node.js is a server-side scripting language and a runtime environment that runs scripts written in various programming languages, including JavaScript and CoffeeScript. This means it can run any type of software with the help of external libraries or frameworks.

Python



Python can run on any operating system as a server-side language, but it's often used for web applications. In fact, many people claim that it is the best programming language to use when developing websites and servers.

Manakah server-side
programming language
kesukaan kalian???

Berikan alasan.



Exercise *for* Students

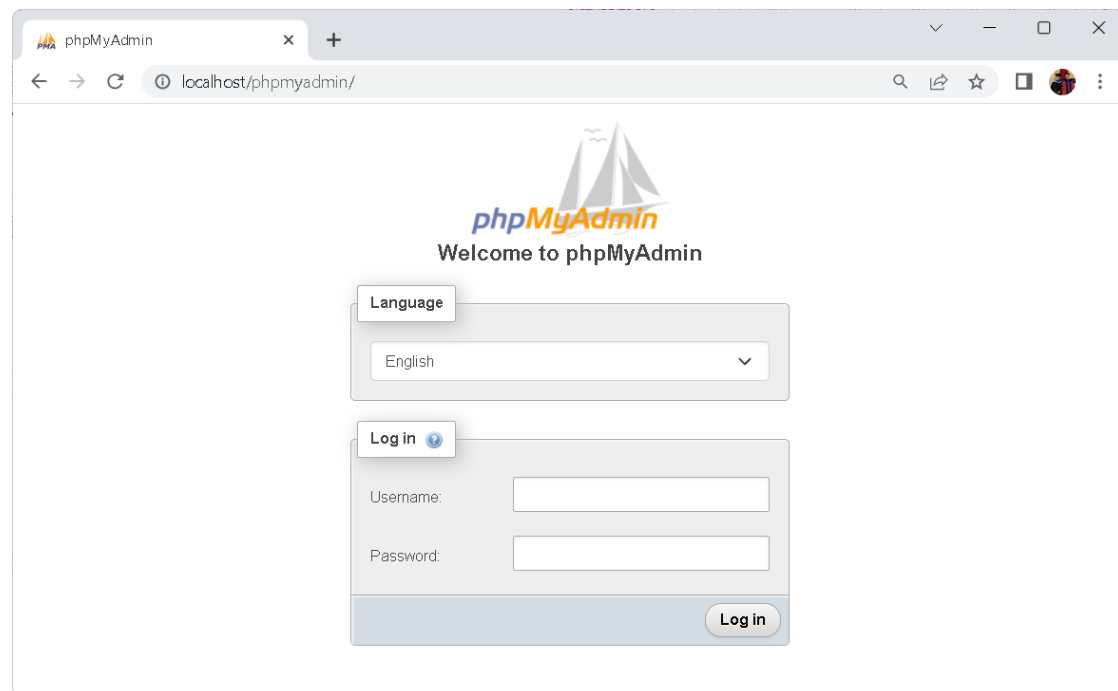
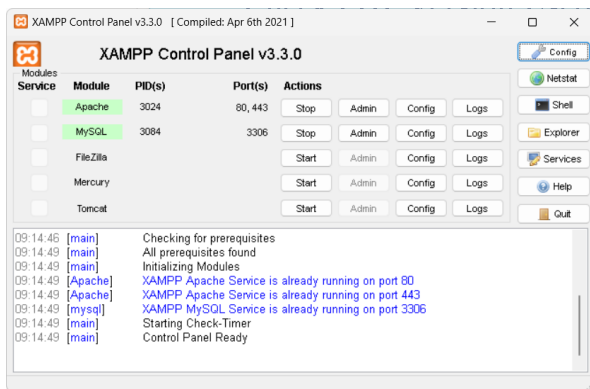
1 – Create *Database* in XAMPP using phpMyAdmin Tools

Xampp Database Management Tools

phpMyAdmin adalah sebuah aplikasi web yang digunakan untuk mengelola dan mengelola database MySQL melalui antarmuka grafis. **phpMyAdmin** dapat digunakan dengan LAMP stack. **LAMP stack** adalah akronim untuk kombinasi *software* yang sering digunakan bersama-sama untuk mengembangkan dan menjalankan aplikasi web, diantaranya yaitu:

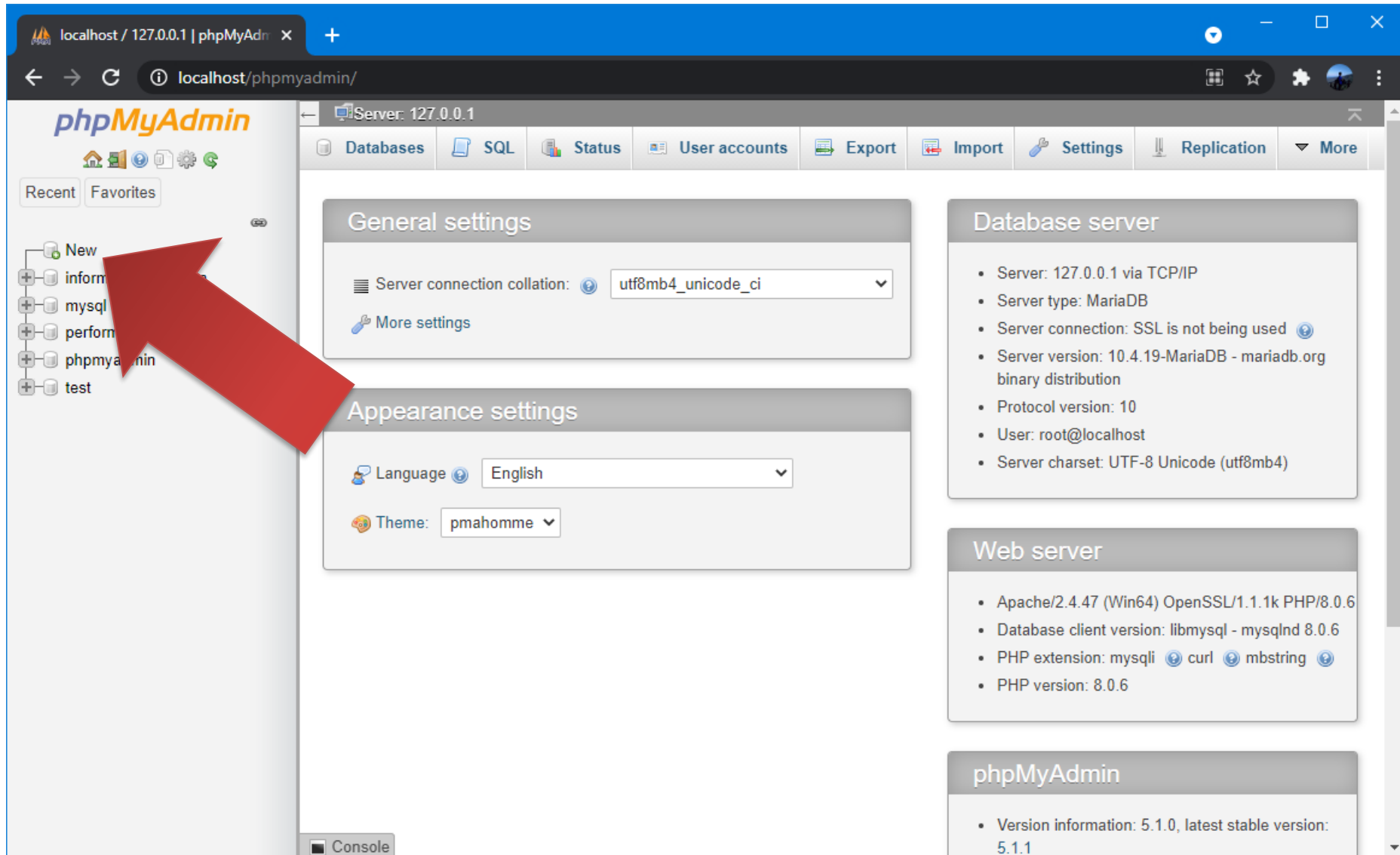
- 1) **Linux**: Sistem operasi berbasis kernel Linux.
- 2) **Apache**: Server web yang umum digunakan untuk menyajikan konten web.
- 3) **MySQL**: Sistem manajemen basis data relasional (RDBMS) yang digunakan untuk menyimpan data.
- 4) **PHP**: Server skrip yang digunakan untuk mengembangkan aplikasi web dinamis.

Link: <http://localhost/phpmyadmin/>



Navigate to Database on phpMyAdmin

Navigate cursor to <http://localhost/phpmyadmin/> and click button “new”.



The screenshot displays the phpMyAdmin web interface in a browser window. The address bar shows the URL `localhost/phpmyadmin/`. The interface includes a top navigation bar with tabs for Databases, SQL, Status, User accounts, Export, Import, Settings, Replication, and More. On the left sidebar, the 'New' button is highlighted with a large red arrow. The main content area is divided into several panels: General settings (Server connection collation: utf8mb4_unicode_ci), Appearance settings (Language: English, Theme: pmahomme), Database server (Server: 127.0.0.1 via TCP/IP, Server type: MariaDB, Server version: 10.4.19-MariaDB, Protocol version: 10, User: root@localhost, Server charset: UTF-8 Unicode (utf8mb4)), Web server (Apache/2.4.47 (Win64) OpenSSL/1.1.1k PHP/8.0.6), and phpMyAdmin (Version information: 5.1.0, latest stable version: 5.1.1).

localhost / 127.0.0.1 | phpMyAdmin

localhost/phpmyadmin/

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Settings Replication More

phpMyAdmin

Recent Favorites

New

inform

mysql

perform

phpmyadmin

test

General settings

Server connection collation: utf8mb4_unicode_ci

More settings

Appearance settings

Language: English

Theme: pmahomme

Database server

- Server: 127.0.0.1 via TCP/IP
- Server type: MariaDB
- Server connection: SSL is not being used
- Server version: 10.4.19-MariaDB - mariadb.org binary distribution
- Protocol version: 10
- User: root@localhost
- Server charset: UTF-8 Unicode (utf8mb4)

Web server

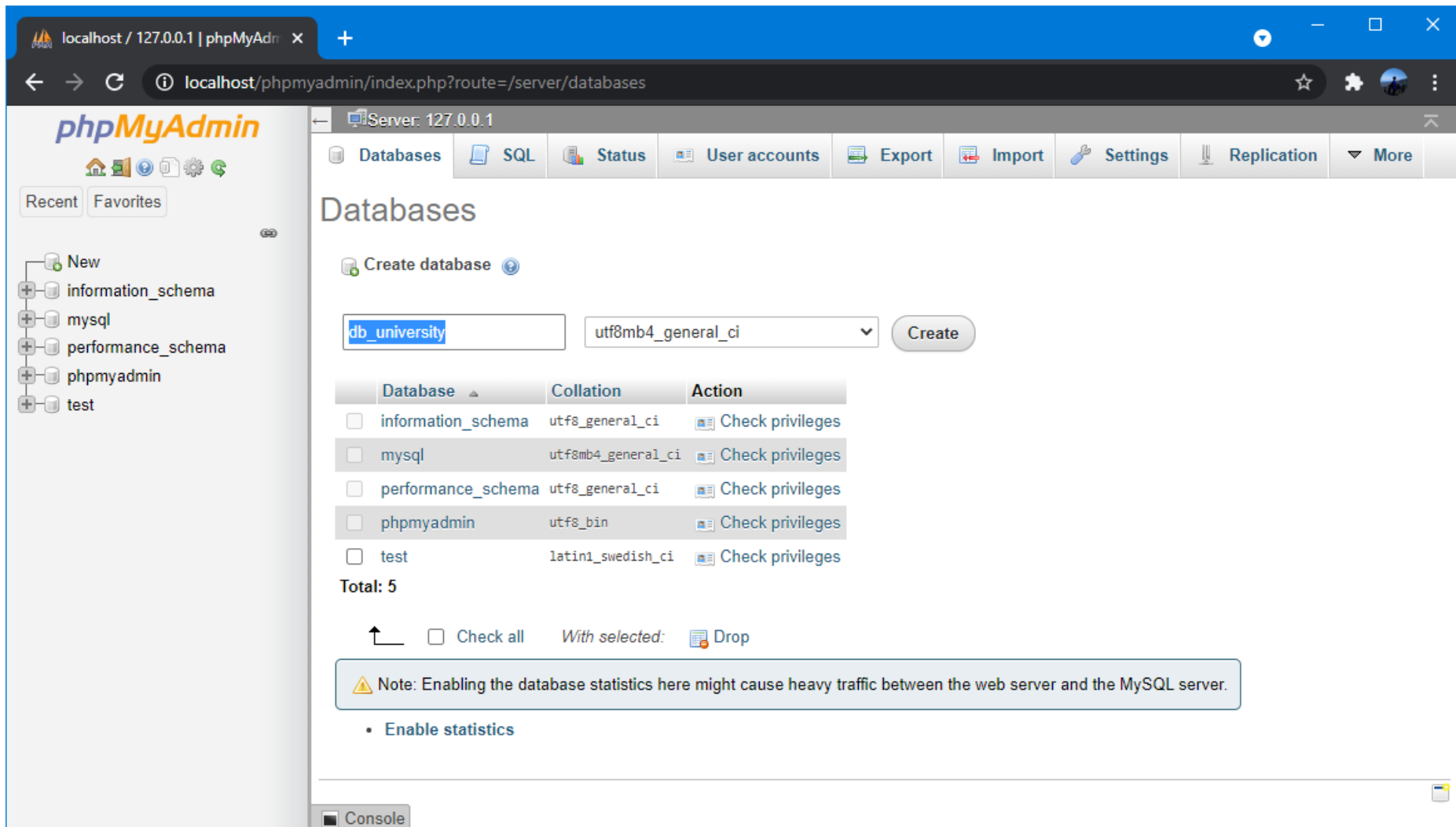
- Apache/2.4.47 (Win64) OpenSSL/1.1.1k PHP/8.0.6
- Database client version: libmysql - mysqlnd 8.0.6
- PHP extension: mysqli curl mbstring
- PHP version: 8.0.6

phpMyAdmin

- Version information: 5.1.0, latest stable version: 5.1.1

Console

Create a Database



The screenshot shows the phpMyAdmin web interface in a browser window. The address bar indicates the URL is `localhost/phpmyadmin/index.php?route=/server/databases`. The interface has a blue header with the phpMyAdmin logo and a navigation menu on the left. The main content area is titled 'Databases' and features a 'Create database' section with an input field containing 'db_university' and a dropdown menu set to 'utf8mb4_general_ci'. A 'Create' button is next to the dropdown. Below this is a table listing existing databases with columns for Database, Collation, and Action.

Database	Collation	Action
☐ information_schema	utf8_general_ci	Check privileges
☐ mysql	utf8mb4_general_ci	Check privileges
☐ performance_schema	utf8_general_ci	Check privileges
☐ phpmyadmin	utf8_bin	Check privileges
☐ test	latin1_swedish_ci	Check privileges

Total: 5

☐ Check all With selected: Drop

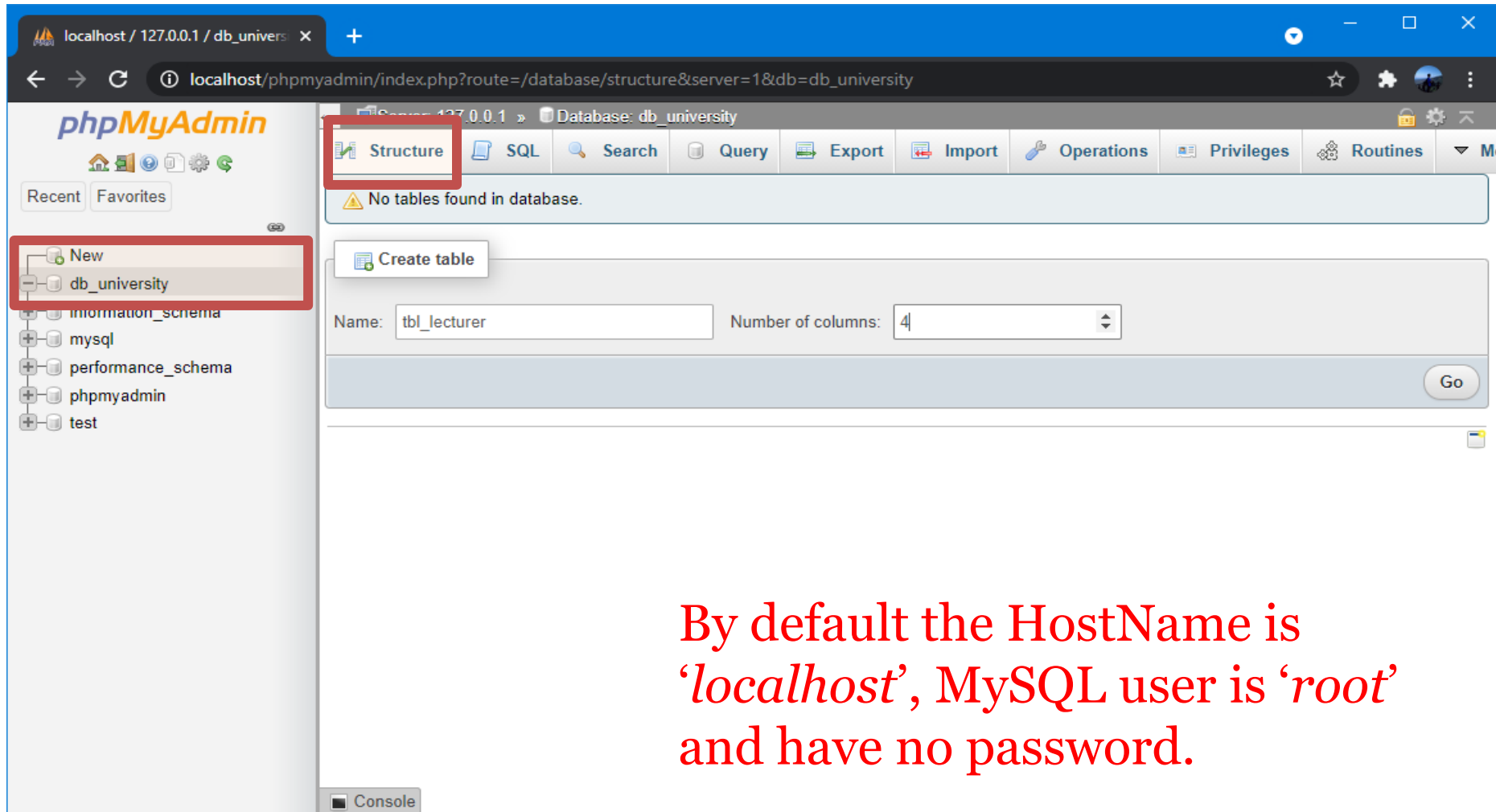
Note: Enabling the database statistics here might cause heavy traffic between the web server and the MySQL server.

- [Enable statistics](#)

Console

See the option to “*Create a Database*” and input field to enter the database name. Write the database name and click the “*Create*” button.

Create Tables Using phpMyAdmin



By default the HostName is 'localhost', MySQL user is 'root' and have no password.

In the phpMyAdmin click on the “Structure” tab and click on the “Create new table on database” wizard to get started. You can see above fields, you can add the **table name** and the total number of fields (*columns*) and click the “Go” button.

Specify Data Type

The screenshot shows the phpMyAdmin interface for the 'db_university' database. The 'tbl_lecturer' table is selected, and the 'Structure' tab is active. The table structure is displayed as follows:

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index
id	VARCHAR	100	None			☐	---
name	VARCHAR	100	None			☐	---
email	VARCHAR	100	None			☐	---
phone	VARCHAR	100	None			☐	---

Below the table structure, the 'Structure' section includes fields for 'Table comments:', 'Collation:', 'Storage Engine:' (set to InnoDB), and 'Connection:'. There is also a 'PARTITION definition:' section with 'Partition by:' and 'Partitions:' fields. At the bottom, there are 'Preview SQL' and 'Save' buttons, and a 'Console' tab is visible.

After the data type, you can specify other optional attributes for each *column*.

Table Structure

The screenshot shows the phpMyAdmin interface for a MySQL database named 'db_university'. The 'Table structure' tab is selected for the table 'tbl_lecturer'. The table has four columns: 'id', 'name', 'email', and 'phone', all of type 'varchar(100)' and collation 'utf8mb4_general_ci'. Below the table structure, there are options to 'Check all', 'With selected', 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', 'Spatial', and 'Fulltext'. There are also buttons for 'Add to central columns' and 'Remove from central columns'. At the bottom, there are sections for 'Indexes' and 'Partitions', both showing 'No index defined!' and 'No partitioning defined!' respectively. The 'Console' tab is visible at the bottom left.

localhost / 127.0.0.1 / db_univers x +

localhost/phpmyadmin/index.php?route=/table/structure&db=db_university&table=tbl_lecturer

Server: 127.0.0.1 » Database: db_university » Table: tbl_lecturer

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Table structure Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	2	name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	3	email	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	4	phone	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More

☐ Check all With selected: [Browse](#) [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Spatial](#) [Fulltext](#)

[Add to central columns](#) [Remove from central columns](#)

[Print](#) [Propose table structure](#) [Track table](#) [Move columns](#) [Normalize](#)

[Add](#) 1 column(s) after phone [Go](#)

Indexes [?](#)

[No index defined!](#)

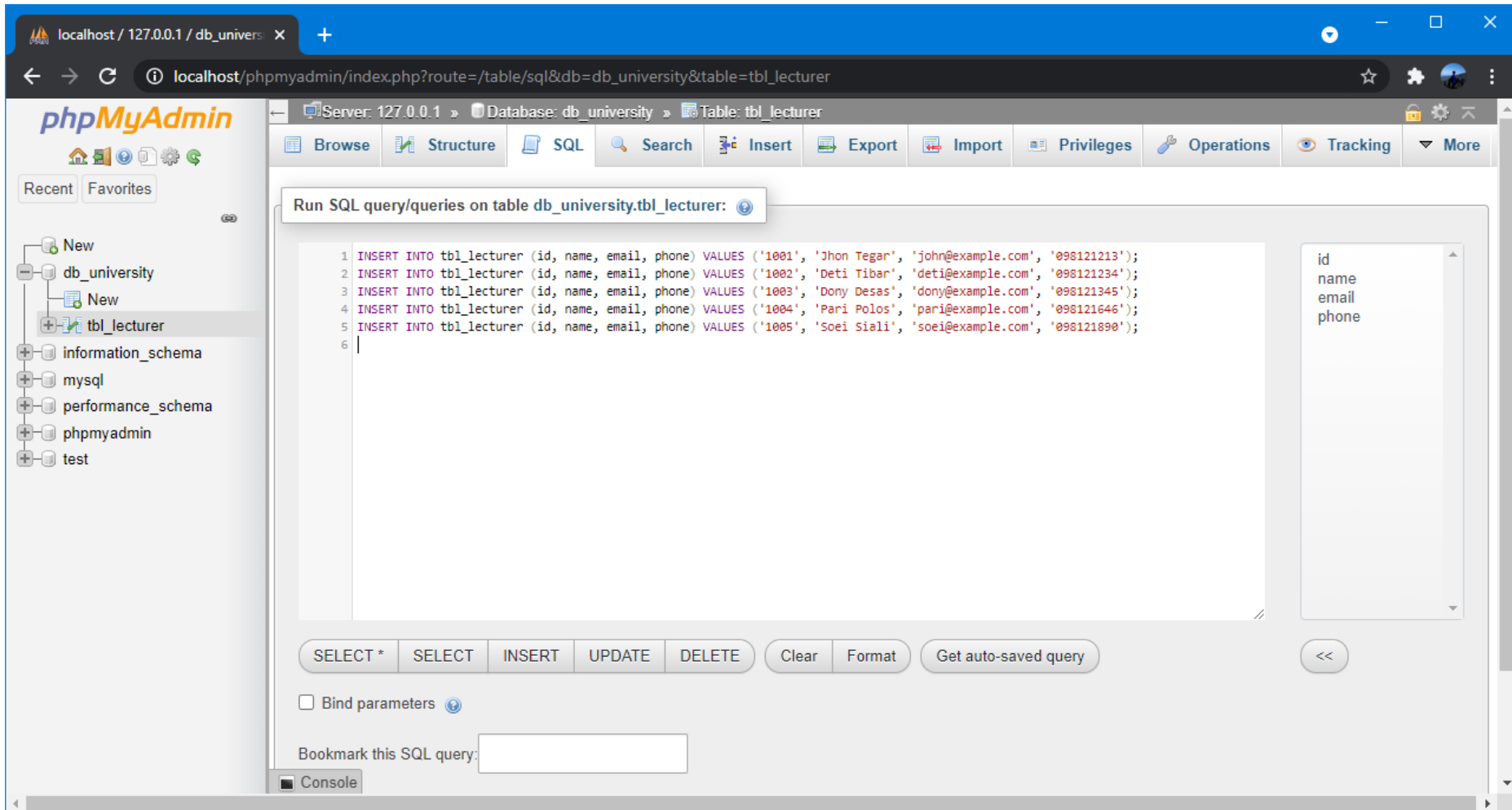
Create an index on 1 columns [Go](#)

Partitions [?](#)

[No partitioning defined!](#)

Console

MySQL Insert Data *via* SQL Suery



The screenshot shows the phpMyAdmin web interface in a browser window. The address bar indicates the URL: `localhost/phpmyadmin/index.php?route=/table/sql&db=db_university&table=tbl_lecturer`. The interface shows the 'db_university' database selected, with the 'tbl_lecturer' table chosen. The 'SQL' tab is active, displaying a query to insert five rows of data into the 'tbl_lecturer' table. The query is as follows:

```
1 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1001', 'Jhon Tegar', 'john@example.com', '098121213');
2 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1002', 'Deti Tibar', 'deti@example.com', '098121234');
3 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1003', 'Dony Desas', 'dony@example.com', '098121345');
4 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1004', 'Pari Polos', 'pari@example.com', '098121646');
5 INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1005', 'Soei Siali', 'soei@example.com', '098121890');
6
```

Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. The 'INSERT' button is highlighted. There is also a checkbox for 'Bind parameters' and a 'Bookmark this SQL query' field.

```
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1001', 'Jhon Tegar', 'john@example.com', '098121213');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1002', 'Deti Tibar', 'deti@example.com', '098121234');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1003', 'Dony Desas', 'dony@example.com', '098121345');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1004', 'Pari Polos', 'pari@example.com', '098121646');
INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('1005', 'Soei Siali', 'soei@example.com', '098121890');
```

Data from a MySQL Database

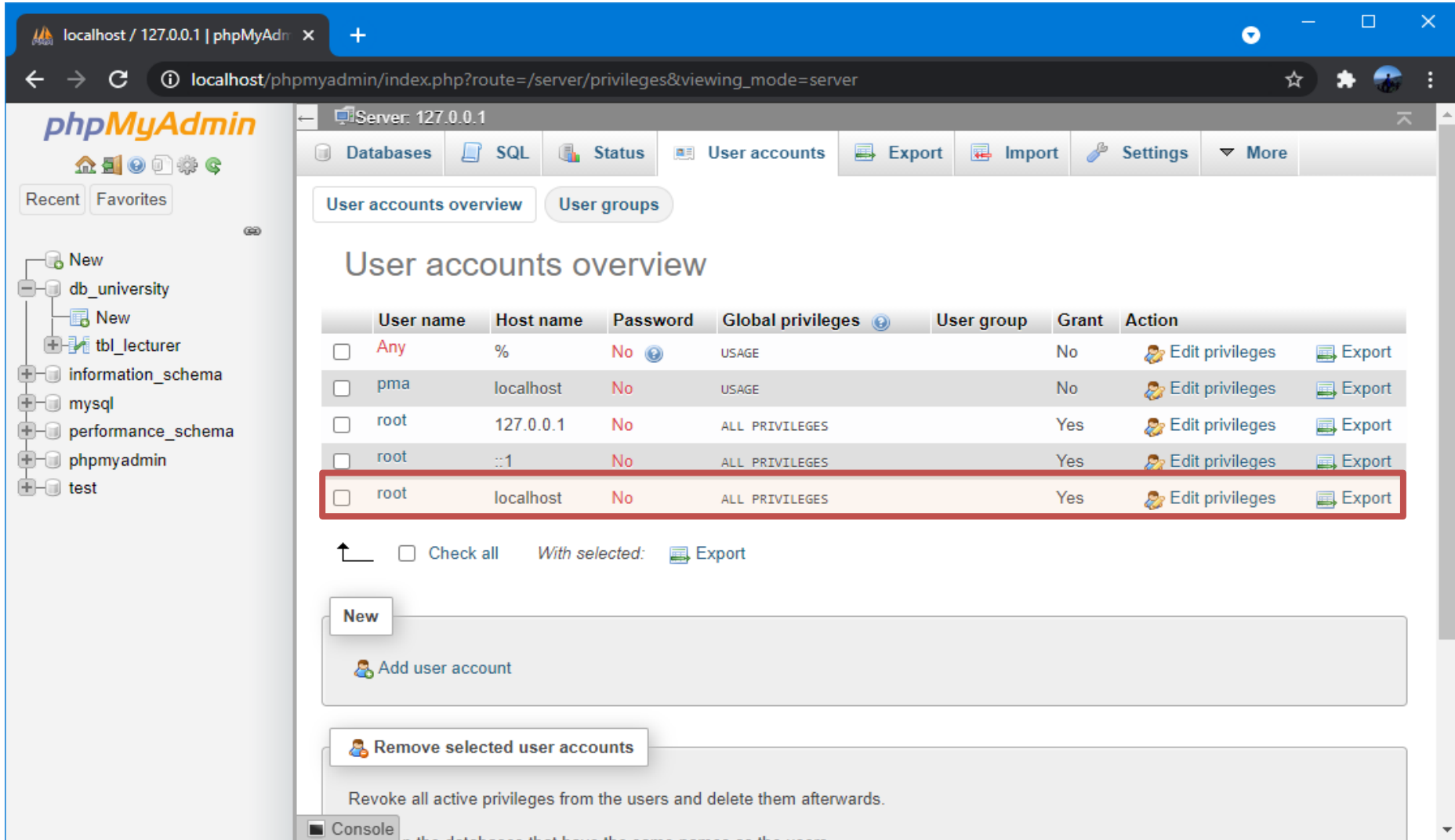
The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL: `localhost/phpmyadmin/index.php?route=/sql&db=db_university&table=tbl_lecturer&pos=0`. The interface is for the `db_university` database, specifically viewing the `tbl_lecturer` table. A warning message states: "Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available." Below this, a green status bar indicates "Showing rows 0 - 4 (5 total, Query took 0.0009 seconds.)". The SQL query entered is `SELECT * FROM `tbl_lecturer``. Below the query, there are options for "Profiling" and links for "Edit inline", "Edit", "Explain SQL", "Create PHP code", and "Refresh". A "Show all" checkbox and a "Number of rows" dropdown set to 25 are visible, along with a "Filter rows" search box. The table data is displayed below the "Options" section. The table has four columns: `id`, `name`, `email`, and `phone`. The data rows are as follows:

id	name	email	phone
1001	Jhon Tegar	john@example.com	098121213
1002	Deti Tibar	deti@example.com	098121234
1003	Dony Desas	dony@example.com	098121345
1004	Pari Polos	pari@example.com	098121646
1005	Soei Siali	soei@example.com	098121890

Below the table, there is another "Show all" checkbox, a "Number of rows" dropdown set to 25, and a "Filter rows" search box. At the bottom, there is a "Query results operations" section with buttons for "Print", "Copy to clipboard", "Export", "Display chart", and "Create view". The "Console" tab is selected at the bottom left.

2 – Set Password to phpMyAdmin *on* XAMPP

User Accounts



The screenshot shows the phpMyAdmin interface for managing user accounts. The browser address bar indicates the URL is `localhost/phpmyadmin/index.php?route=/server/privileges&viewing_mode=server`. The left sidebar shows a tree view of databases, including `db_university`, `tbl_lecturer`, `information_schema`, `mysql`, `performance_schema`, `phpmyadmin`, and `test`. The main content area is titled "User accounts overview" and displays a table of user accounts. The table has columns for "User name", "Host name", "Password", "Global privileges", "User group", "Grant", and "Action". The row for the user "root" with host "localhost" is highlighted with a red border. Below the table, there are buttons for "New" (Add user account) and "Remove selected user accounts".

	User name	Host name	Password	Global privileges	User group	Grant	Action
☐	Any	%	No	USAGE		No	Edit privileges Export
☐	pma	localhost	No	USAGE		No	Edit privileges Export
☐	root	127.0.0.1	No	ALL PRIVILEGES		Yes	Edit privileges Export
☐	root	::1	No	ALL PRIVILEGES		Yes	Edit privileges Export
☐	root	localhost	No	ALL PRIVILEGES		Yes	Edit privileges Export

Below the table, there are buttons for "New" (Add user account) and "Remove selected user accounts".

Click on the “user accounts” option at the top of the page and press the “*edit privileges*” under “actions” option for the Username “*root*” and Hostname “*localhost*”.

Change/Edit Password

The screenshot shows the phpMyAdmin web interface in a browser window. The address bar displays the URL: `localhost/phpmyadmin/index.php?route=/server/privileges&username=root&hostname=localhost&dbname=&tablename=&...`. The interface has a blue header with the phpMyAdmin logo and a navigation bar with tabs: Databases, SQL, Status, User accounts, Export, Import, Settings, Replication, and More. Below the navigation bar, there are sub-tabs: Global, Database, Change password, and Login Information. The 'Change password' tab is selected. The main content area is titled 'Edit privileges: User account 'root'@'localhost'' and contains a warning note: 'Note: You are attempting to edit privileges of the user with which you are currently logged in.' Below the note, there is a 'Change password' section with the following options: 'No Password' (radio button), 'Password' (radio button, selected), and 'Password Hashing' (dropdown menu set to 'Native MySQL authentication'). There are input fields for 'Enter:' and 'Re-type:' passwords, both masked with dots. A 'Strength' indicator shows a progress bar and the text 'Extremely weak'. At the bottom of the section, there is a 'Generate password' button and a text input field. A 'Go' button is located at the bottom right of the form area. The left sidebar shows a tree view of the database structure, including 'db_university', 'tbl_lecturer', 'information_schema', 'mysql', 'performance_schema', 'phpmyadmin', and 'test'.

localhost / 127.0.0.1 | phpMyAdmin

localhost/phpmyadmin/index.php?route=/server/privileges&username=root&hostname=localhost&dbname=&tablename=&...

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Settings Replication More

Global Database Change password Login Information

Edit privileges: User account 'root'@'localhost'

Note: You are attempting to edit privileges of the user with which you are currently logged in.

Change password

☐ No Password

☒ Password:

Enter: Strength: Extremely weak

Re-type:

Password Hashing: Native MySQL authentication

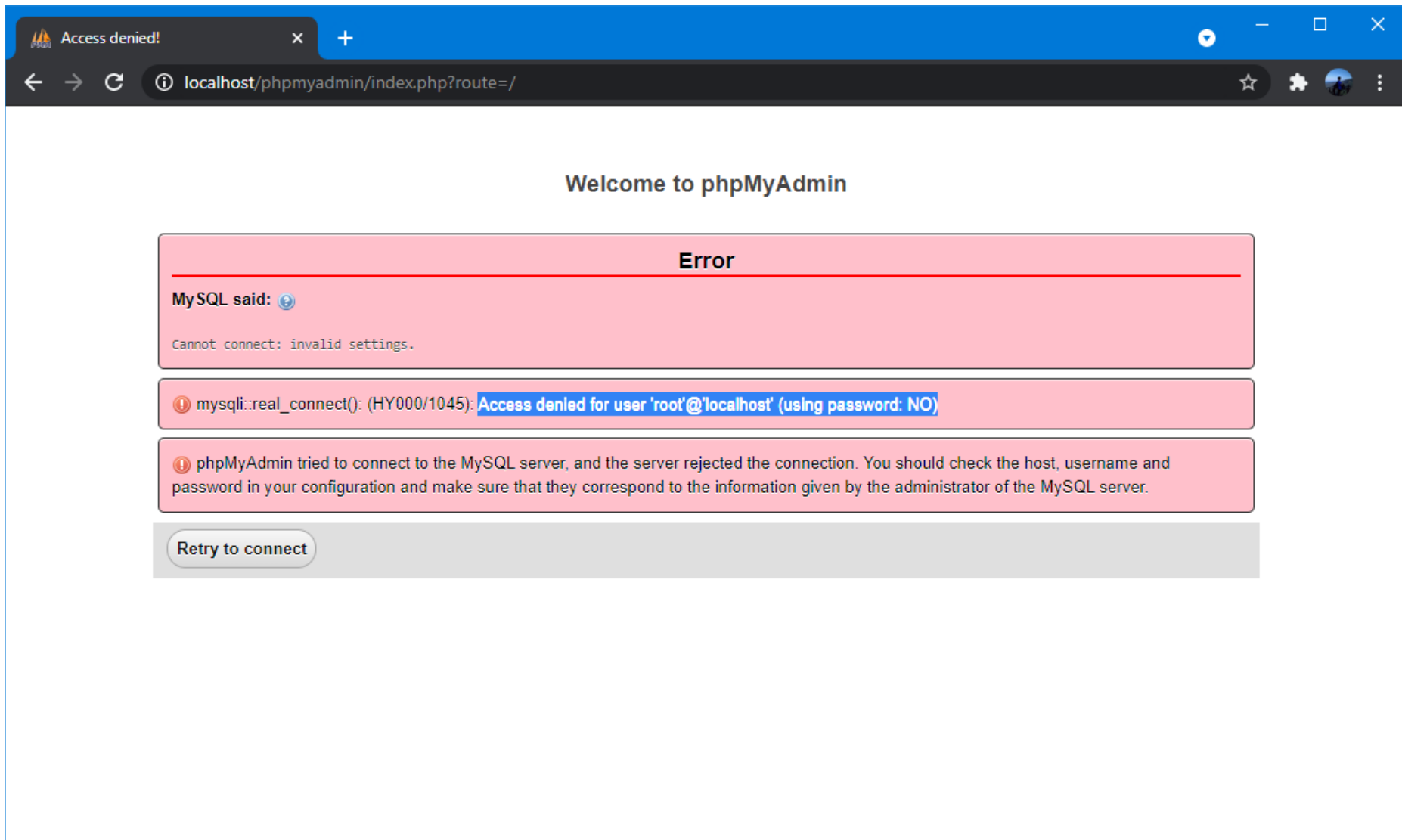
Generate password Generate

Go

Console

Choose the third tab “*change password*” and type your password in the provided field, retype the password to confirm it and then finally click on the “Go” button to conclude the process.

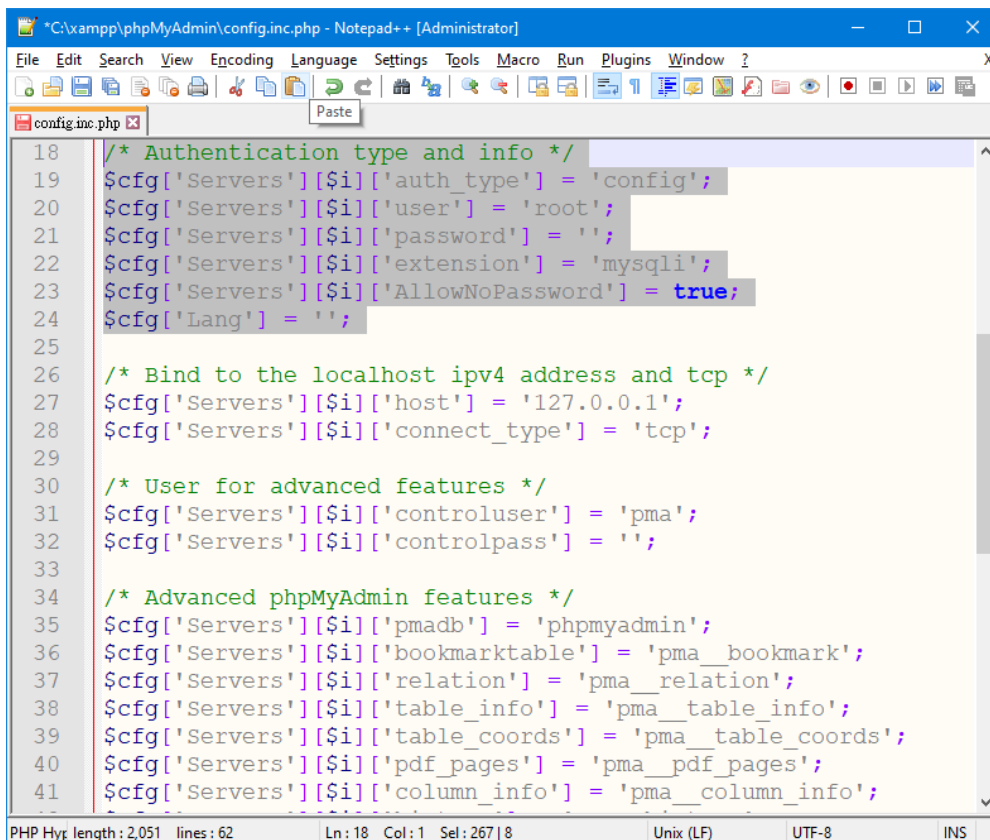
Error Access Denied



Configuration Settings [*allow without password*]

ERROR #1045 - Access denied for user 'root'@'localhost' (using password: NO) when opening **http://localhost/phpmyadmin**.

- ❖ Navigate to "**C:\xampp\phpMyAdmin**"
- ❖ Find the file "**config.inc.php**" and open in a Notepad or better, Notepad++ (**run as administrator**).
- ❖ At line # 18 to 24, is the settings for *user authentication*.



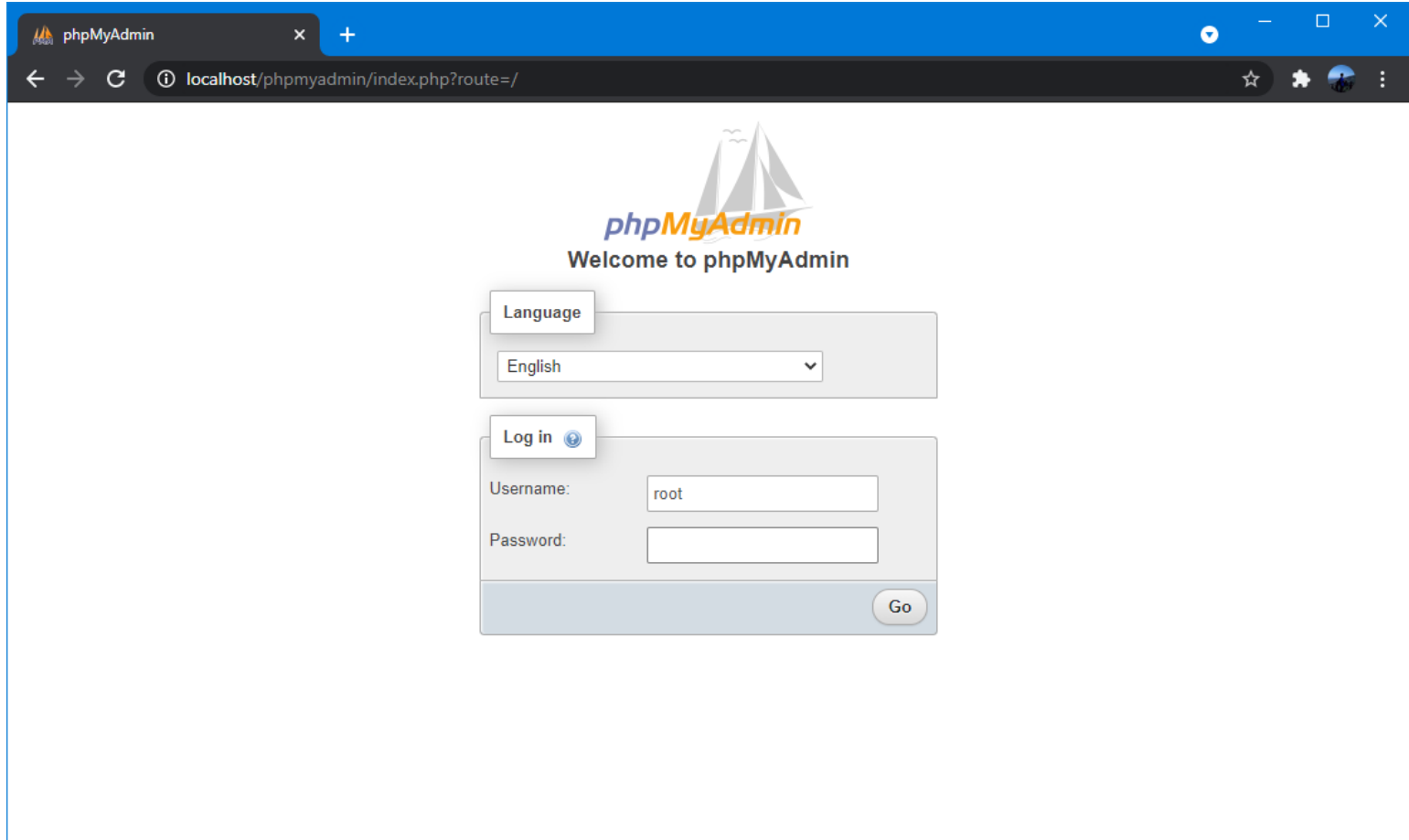
```
18  /* Authentication type and info */
19  $cfg['Servers'][$i]['auth_type'] = 'config';
20  $cfg['Servers'][$i]['user'] = 'root';
21  $cfg['Servers'][$i]['password'] = '';
22  $cfg['Servers'][$i]['extension'] = 'mysqli';
23  $cfg['Servers'][$i]['AllowNoPassword'] = true;
24  $cfg['Lang'] = '';
25
26  /* Bind to the localhost ipv4 address and tcp */
27  $cfg['Servers'][$i]['host'] = '127.0.0.1';
28  $cfg['Servers'][$i]['connect_type'] = 'tcp';
29
30  /* User for advanced features */
31  $cfg['Servers'][$i]['controluser'] = 'pma';
32  $cfg['Servers'][$i]['controlpass'] = '';
33
34  /* Advanced phpMyAdmin features */
35  $cfg['Servers'][$i]['pmadb'] = 'phpmyadmin';
36  $cfg['Servers'][$i]['bookmarktable'] = 'pma_bookmark';
37  $cfg['Servers'][$i]['relation'] = 'pma_relation';
38  $cfg['Servers'][$i]['table_info'] = 'pma_table_info';
39  $cfg['Servers'][$i]['table_coords'] = 'pma_table_coords';
40  $cfg['Servers'][$i]['pdf_pages'] = 'pma_pdf_pages';
41  $cfg['Servers'][$i]['column_info'] = 'pma_column_info';
```

Change the values to :

```
/* Authentication type and info */
$cfg['Servers'][$i]['auth_type'] = 'cookie';
$cfg['Servers'][$i]['user'] = 'root';
$cfg['Servers'][$i]['password'] = '12345';
$cfg['Servers'][$i]['extension'] = 'mysqli';
$cfg['Servers'][$i]['AllowNoPassword'] = false;
$cfg['Lang'] = '';
```

password: password is not set in the configuration file.

Sign In Testing



A screenshot of a web browser displaying the phpMyAdmin login interface. The browser's address bar shows the URL `localhost/phpmyadmin/index.php?route=/'`. The page features the phpMyAdmin logo, which includes a stylized sailboat, and the text "Welcome to phpMyAdmin". Below the logo, there is a "Language" dropdown menu currently set to "English". Underneath, a "Log in" button is visible. The login form contains two input fields: "Username:" with the text "root" entered, and "Password:" which is empty. A "Go" button is located at the bottom right of the form.

phpMyAdmin
Welcome to phpMyAdmin

Language
English

Log in

Username: root

Password:

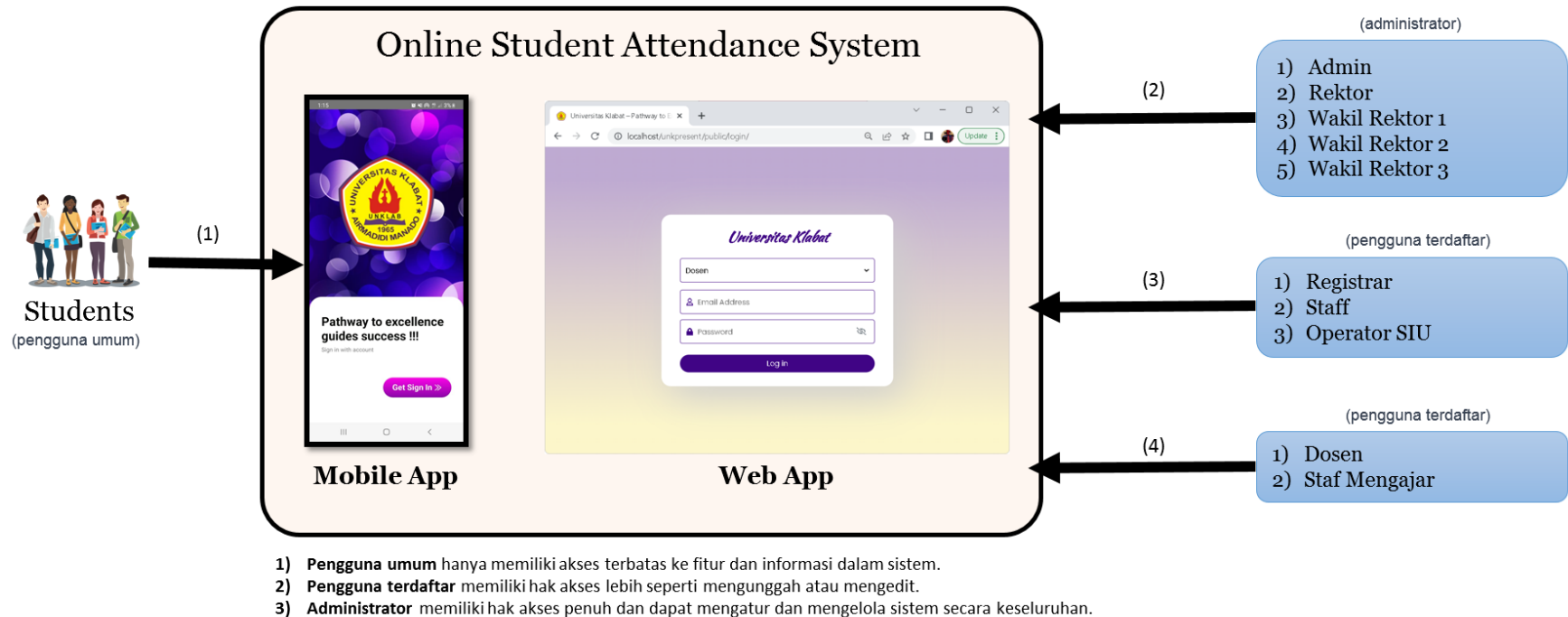
Go

Reset Root Password

In case when you have installed XAMPP, you did the security settings, but after a time you have forgot the *root* password.

1. Open folder “**C:/xampp/mysql/**”
> Run this batch file: **resetroot.bat**
2. Go to the “**C:/xampp/mysql/bin/**” folder.
3. Edit **my.ini** and insert ***skip-grant-tables*** below [mysqld].
4. Restart MySQL (*Open xampp control panel and stop it*).
5. Buka mysql:
> **mysql -u root -p**
> **password:** (*password kosong di enter saja*)
6. Set new password for the *root user* by running **UPDATE mysql.user SET Password=PASSWORD('new_password') WHERE User='root'** in phpMyAdmin in the MySQL database.
7. Force the password prompt for the *root user*:
> FLUSH PRIVILEGES;
> ALTER USER root@localhost IDENTIFIED VIA mysql_native_password USING PASSWORD('new secret password');
> FLUSH PRIVILEGES;
8. Don't forget to edit “**my.ini**” again and remove ***skip-grant-tables*** below [mysqld].

3 – Create More tables



Sistem absensi mahasiswa secara online dengan dua aplikasi (*mobile app dan web app*) memiliki tujuan untuk memudahkan *students* dan *staf* dalam melakukan absensi dan pengelolaan kehadiran mahasiswa di kelas.

Nama table: tbl_operator

Database Name: “db_attendance”

Table Name: *tbl_operator*

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	nip	varchar(150)	utf8mb4_0900_ai_ci		No	None			 Change  Drop  More
☐ 2	email	varchar(50)	utf8mb4_0900_ai_ci		No	None			 Change  Drop  More
☐ 3	fullname	varchar(100)	utf8mb4_0900_ai_ci		No	None			 Change  Drop  More
☐ 4	password	varchar(100)	utf8mb4_0900_ai_ci		No	None			 Change  Drop  More
☐ 5	phone_number	varchar(50)	utf8mb4_0900_ai_ci		No	None			 Change  Drop  More
☐ 6	role	varchar(100)	utf8mb4_0900_ai_ci		No	None			 Change  Drop  More
☐ 7	created_at	varchar(150)	utf8mb4_0900_ai_ci		No	None			 Change  Drop  More
☐ 8	updated_at	varchar(150)	utf8mb4_0900_ai_ci		No	None			 Change  Drop  More

```
INSERT INTO `tbl_operator` (`nip`, `email`, `fullname`, `password`, `phone_number`, `role`, `created_at`, `updated_at`) VALUES ('23456', 'stenly@unklab.ac.id', 'Stenly Adam', '12345', '0852123456', 'admin', '12-04-2023 12:04:32', '12-04-2023 12:04:32');
```

```
INSERT INTO `tbl_operator` (`nip`, `email`, `fullname`, `password`, `phone_number`, `role`, `created_at`, `updated_at`) VALUES ('34567', 'reymond@unklab.ac.id', 'Reymond Rotikan', '12345', '0852123456', 'staff', '12-04-2023 12:04:32', '12-04-2023 12:04:32');
```

```
INSERT INTO `tbl_operator` (`nip`, `email`, `fullname`, `password`, `phone_number`, `role`, `created_at`, `updated_at`) VALUES ('12345', 'andrew@unklab.ac.id', 'Andrew Liem', 'abc123', '0852123456', 'admin', '12-04-2023 12:04:32', '12-04-2023 12:04:32');
```

```
INSERT INTO `tbl_operator` (`nip`, `email`, `fullname`, `password`, `phone_number`, `role`, `created_at`, `updated_at`) VALUES ('12345', 'gladis@unklab.ac.id', 'Gladis Sepang', 'abc123', '0852123456', 'admin', '12-04-2023 12:04:32', '12-04-2023 12:04:32');
```

Nama table: tbl_students

Database Name: “db_attendance”

Table Name: *tbl_students*

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	reg_number	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	2	nim_number	varchar(80)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	3	email	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	4	fullname	varchar(200)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	5	password	varchar(50)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	6	created_at	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	7	updated_at	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More

```
INSERT INTO `tbl_students` (`reg_number`,`nim_number`,`email`,`fullname`,`password`,`created_at`,`updated_at`) VALUES ('123456789012','954321','s954321@student.unklab.ac.id','Putri Moningga','12345','2023-04-13 22:54:33','2023-04-13 22:54:33');
INSERT INTO `tbl_students` (`reg_number`,`nim_number`,`email`,`fullname`,`password`,`created_at`,`updated_at`) VALUES ('123456789012','854321','s854321@student.unklab.ac.id','Sandra Dewi','12345','2023-04-13 22:54:33','2023-04-13 22:54:33');
INSERT INTO `tbl_students` (`reg_number`,`nim_number`,`email`,`fullname`,`password`,`created_at`,`updated_at`) VALUES ('123456789012','754321','s754321@student.unklab.ac.id','Citra Lentari Anada','12345','2023-04-13 22:54:33','2023-04-13 22:54:33');
INSERT INTO `tbl_students` (`reg_number`,`nim_number`,`email`,`fullname`,`password`,`created_at`,`updated_at`) VALUES ('123456789012','554321','s554321@student.unklab.ac.id','Dinda Manopo Putri','12345','2023-04-13 22:54:33','2023-04-13 22:54:33');
INSERT INTO `tbl_students` (`reg_number`,`nim_number`,`email`,`fullname`,`password`,`created_at`,`updated_at`) VALUES ('123456789012','454321','s454321@student.unklab.ac.id','Miehie Kawekiang Sie','12345','2023-04-13 22:54:33','2023-04-13 22:54:33');
INSERT INTO `tbl_students` (`reg_number`,`nim_number`,`email`,`fullname`,`password`,`created_at`,`updated_at`) VALUES ('123456789012','354321','s354321@student.unklab.ac.id','ChinChin Long','12345','2023-04-13 22:54:33','2023-04-13 22:54:33');
INSERT INTO `tbl_students` (`reg_number`,`nim_number`,`email`,`fullname`,`password`,`created_at`,`updated_at`) VALUES ('123456789012','254321','s254321@student.unklab.ac.id','Putra Podomi','12345','2023-04-13 22:54:33','2023-04-13 22:54:33');
INSERT INTO `tbl_students` (`reg_number`,`nim_number`,`email`,`fullname`,`password`,`created_at`,`updated_at`) VALUES ('123456789012','154321','s154321@student.unklab.ac.id','Radix Basalama','12345','2023-04-13 22:54:33','2023-04-13 22:54:33');
INSERT INTO `tbl_students` (`reg_number`,`nim_number`,`email`,`fullname`,`password`,`created_at`,`updated_at`) VALUES ('123456789012','604321','s604321@student.unklab.ac.id','Margono Raharjo','12345','2023-04-13 22:54:33','2023-04-13 22:54:33');
INSERT INTO `tbl_students` (`reg_number`,`nim_number`,`email`,`fullname`,`password`,`created_at`,`updated_at`) VALUES ('123456789012','644321','s644321@student.unklab.ac.id','Sempoyonangan Ojomiso','12345','2023-04-13 22:54:33','2023-04-13 22:54:33');
```

Nama table: tbl_classes

Database Name: “db_attendance”

Table Name: *tbl_classes*

Table Lecturer Class:

- 1) Unik ID Subject/Class/Matakuliah
- 2) (Otomatis) Lecturer Email
- 3) Code Matakuliah
- 4) Nama Subject/Class/Matakuliah
- 5) Lecturer Name (Pengajar)
- 6) Fakultas
- 7) Jurusan (Program Studi)
- 8) SKS
- 9) Building & Nomor Ruangan (Room)
- 10) Room Latitude
- 11) Room Longitude
- 12) Jadwal Class Hari/Waktu (Day/Time)
- 13) Daftar Email Mahasiswa
- 14) Created_at
- 15) **Status Class (active or completed)**

Columns:

- 1) `id\_class`, **(Primary Key)**
- 2) `email\_lecturer`,
- 3) `code\_class`,
- 4) `name\_subject`,
- 5) `name\_lecturer`,
- 6) `fakultas`,
- 7) `prodi`,
- 8) `sks`,
- 9) `building\_room`,
- 10) `room\_latitude`,
- 11) `room\_longitude`,
- 12) `jadwal\_class\_day\_time`,
- 13) `daftar\_email\_student`, (Data type: Text)
- 14) `created\_at`,
- 15) `status\_class`

Nama table: tbl_attendance_list

Database Name: “db_attendance”

Table Name: *tbl_attendance_list*

Table Daftar Schedule Attendance:

- 1) Unik ID schedule presensi untuk setiap daftar kehadiran Class
- 2) Title/Judul/Deskripsi yang unik untuk pengisian presensi
- 3) Jadwal *date* pengisian presensi (*tanggal pengambilan kehadiran*)
- 4) Jadwal *time* pengisian presensi (*waktu pengambilan kehadiran*)
- 5) **Unik ID Subject/Class/Matakuliah**
- 6) Nama Subject/Class/Matakuliah
- 7) Lecturer Email
- 8) Lecturer Name
- 9) Room Latitude
- 10) Room Longitude
- 11) **Maximum radius (*jarak/distance/zone*) pengambilan presensi**
- 12) Created_at

Columns:

- 1) `id\_attendance`,
- 2) `title\_short`,
- 3) `date\_attendance`,
- 4) `time\_attendance`,
- 5) `id\_class`,
- 6) `name\_subject`,
- 7) `email\_lecturer`,
- 8) `name\_lecturer`,
- 9) `room\_latitude`,
- 10) `room\_longitude`,
- 11) `max\_radius`,
- 12) `created\_at`

Nama table: tbl_attendance_history

Database Name: “db_attendance”

Table Name: *tbl_attendance_history*

Table History Kehadiran Mahasiswa:

- 1) ID Unik Subject/class/Matakuliah
- 2) ID Unik attendance schedule untuk setiap daftar kehadiran Class
- 3) Student Email
- 4) Nama Matakuliah
- 5) Lecturer Email
- 6) Longitude Posisi Student (*Bukan koordinat class*)
- 7) Latitude Posisi Student (*Bukan koordinat class*)
- 8) Jarak/distance pada saat pengambilan presensi
- 9) Selesai Dalam waktu
- 10) Status Kehadiran (Hadir, Late, Absen, Izin)
- 11) Note (*keterangan sakit, izin, etc.*)
- 12) Created at (*tanggal/waktu pada saat pengambilan absen*)

Columns:

- 1) `id\_class`,
- 2) `id\_attendance`,
- 3) `email\_student`,
- 4) `name\_subject`,
- 5) `email\_lecturer`,
- 6) `student\_lat`,
- 7) `student\_long`,
- 8) `distance`,
- 9) `time\_take\_attendance`,
- 10) `status`,
- 11) `note`,
- 12) `created\_at`

```
INSERT INTO 'tbl_attendance_history' ('id_class', 'id_attendance', 'email_student', 'name_subject', 'email_lecturer', 'student_lat', 'student_long', 'distance', 'time_take_attendance', 'status', 'note', 'created_at') VALUES ('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's154321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '16', '10:00:49', 'P', '', '2023-04-13 22:54:33');
```

```
INSERT INTO 'tbl_attendance_history' ('id_class', 'id_attendance', 'email_student', 'name_subject', 'email_lecturer', 'student_lat', 'student_long', 'distance', 'time_take_attendance', 'status', 'note', 'created_at') VALUES ('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's254321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '12', '10:05:49', 'P', '', '2023-04-13 22:54:33');
```

```
INSERT INTO 'tbl_attendance_history' ('id_class', 'id_attendance', 'email_student', 'name_subject', 'email_lecturer', 'student_lat', 'student_long', 'distance', 'time_take_attendance', 'status', 'note', 'created_at') VALUES ('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's354321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '20', '10:08:49', 'P', '', '2023-04-13 22:54:33');
```

```
INSERT INTO 'tbl_attendance_history' ('id_class', 'id_attendance', 'email_student', 'name_subject', 'email_lecturer', 'student_lat', 'student_long', 'distance', 'time_take_attendance', 'status', 'note', 'created_at') VALUES ('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's454321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '15', '10:20:49', 'P', '', '2023-04-13 22:54:33');
```

```
INSERT INTO 'tbl_attendance_history' ('id_class', 'id_attendance', 'email_student', 'name_subject', 'email_lecturer', 'student_lat', 'student_long', 'distance', 'time_take_attendance', 'status', 'note', 'created_at') VALUES ('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's554321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '6', '10:30:49', 'P', '', '2023-04-13 22:54:33');
```

```
INSERT INTO 'tbl_attendance_history' ('id_class', 'id_attendance', 'email_student', 'name_subject', 'email_lecturer', 'student_lat', 'student_long', 'distance', 'time_take_attendance', 'status', 'note', 'created_at') VALUES ('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's644321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '7', '10:45:49', 'P', '', '2023-04-13 22:54:33');
```

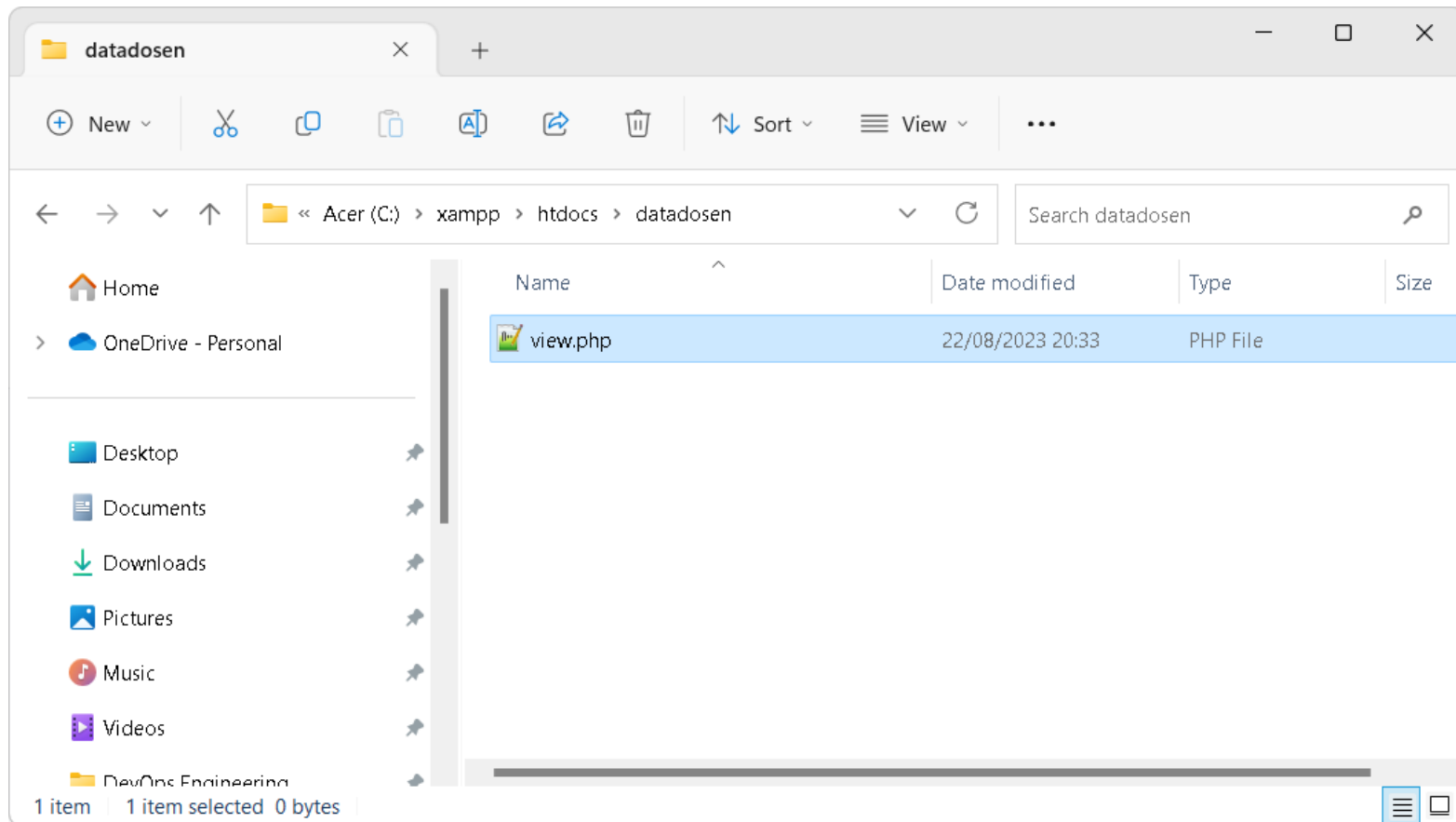
```
INSERT INTO 'tbl_attendance_history' ('id_class', 'id_attendance', 'email_student', 'name_subject', 'email_lecturer', 'student_lat', 'student_long', 'distance', 'time_take_attendance', 'status', 'note', 'created_at') VALUES ('CLASS643ac4d56e4a8', 'ATD745ac4d56e4b7', 's654321@student.unklab.ac.id', 'Artificial Intelligence', 'semmy@unklab.ac.id', '1,239484', '124,348583', '8', '10:27:49', 'P', '', '2023-04-13 22:54:33');
```

4 – View Data in Table *tbl_lecturer* using Server Script Language

Create Working Directory

Working Directory:

C:/xampp/htdocs/datadosen



Xampp "htdocs" Directory

- ❖ Dalam XAMPP, direktori "**htdocs**" memiliki peran penting sebagai **direktori utama** tempat kita menyimpan file-file proyek web. "**htdocs**" adalah singkatan dari "HyperText Documents," folder ini adalah tempat di mana kita dapat menyimpan file HTML, CSS, JavaScript, gambar, dan file-file lain yang berkaitan dengan project web.
- ❖ Ketika mengakses "**localhost**" atau "**127.0.0.1**" pada browser web setelah mengaktifkan server Apache di XAMPP, server web akan merujuk ke direktori "*htdocs*" sebagai *root* dari server lokal. Ini berarti semua file dan folder yang diletakkan di dalam direktori "htdocs" dapat diakses melalui server local (**public directory**).



Jika kita memiliki file dengan nama "index.html" di dalam direktori "htdocs," kita dapat mengaksesnya di folder melalui web browser dengan menavigasi ke "http://localhost/index.html."

Write PHP & HTML Codes

```
C:\xampp\htdocs\datadosen\view.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
view.php x
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>List</title>
7 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
8 </head>
9 <body>
10 <div class="container mt-5"> <h2>Daftar Dosen</h2>
11 <table class="table table-bordered">
12 <thead>
13 <tr>
14 <th>ID</th>
15 <th>Name</th>
16 <th>Email</th>
17 <th>Phone</th>
18 </tr>
19 </thead>
20 <tbody>
21 <!-- PHP CODE HERE -->
22 </tbody>
23 </table>
24 </div>
25 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
26 </body>
27 </html>
28
29
30
31
32
33
34
35
36
PHP Hypertext Preprocessor file length: 914 lines: 37 Ln: 1 Col: 1 Pos: 1 Windows (CR LF) UTF-8 INS
```

```
<tbody>
<?php
$servername = "localhost"; // Database connection
$username = "root"; // User database
$password = "12345"; // Password database
$dbname = "db_university"; // Database name
$conn = new mysqli($servername, $username, $password, $dbname); // connect

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// Fetch data from tbl_lecturer
$result = $conn->query("SELECT * FROM tbl_lecturer"); // Sql query

if ($result->num_rows > 0) {
    while ($row = $result->fetch_assoc()) {
        echo "<tr>";
        echo "<td>" . $row["id"] . "</td>";
        echo "<td>" . $row["name"] . "</td>";
        echo "<td>" . $row["email"] . "</td>";
        echo "<td>" . $row["phone"] . "</td>";
        echo "</tr>";
    }
} else {
    echo "<tr><td colspan='4'>No data found</td></tr>";
}

$conn->close();
?>
</tbody>
```

```
<tbody>
```

```
<?php
```

```
$servername = "localhost"; // Database connection
$username = "root"; // User database
$password = "12345"; // Password database
$dbname = "db_university"; // Database name
$conn = new mysqli($servername, $username, $password, $dbname); // connect
```

```
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
```

```
// Fetch data from tbl_lecturer
$result = $conn->query("SELECT * FROM tbl_lecturer"); // Sql query
```

```
if ($result->num_rows > 0) {
    while ($row = $result->fetch_assoc()) {
        echo "<tr>";
        echo "<td>" . $row["id"] . "</td>";
        echo "<td>" . $row["name"] . "</td>";
        echo "<td>" . $row["email"] . "</td>";
        echo "<td>" . $row["phone"] . "</td>";
        echo "</tr>";
    }
} else {
    echo "<tr><td colspan='4'>No data found</td></tr>";
}
```

```
$conn->close();
```

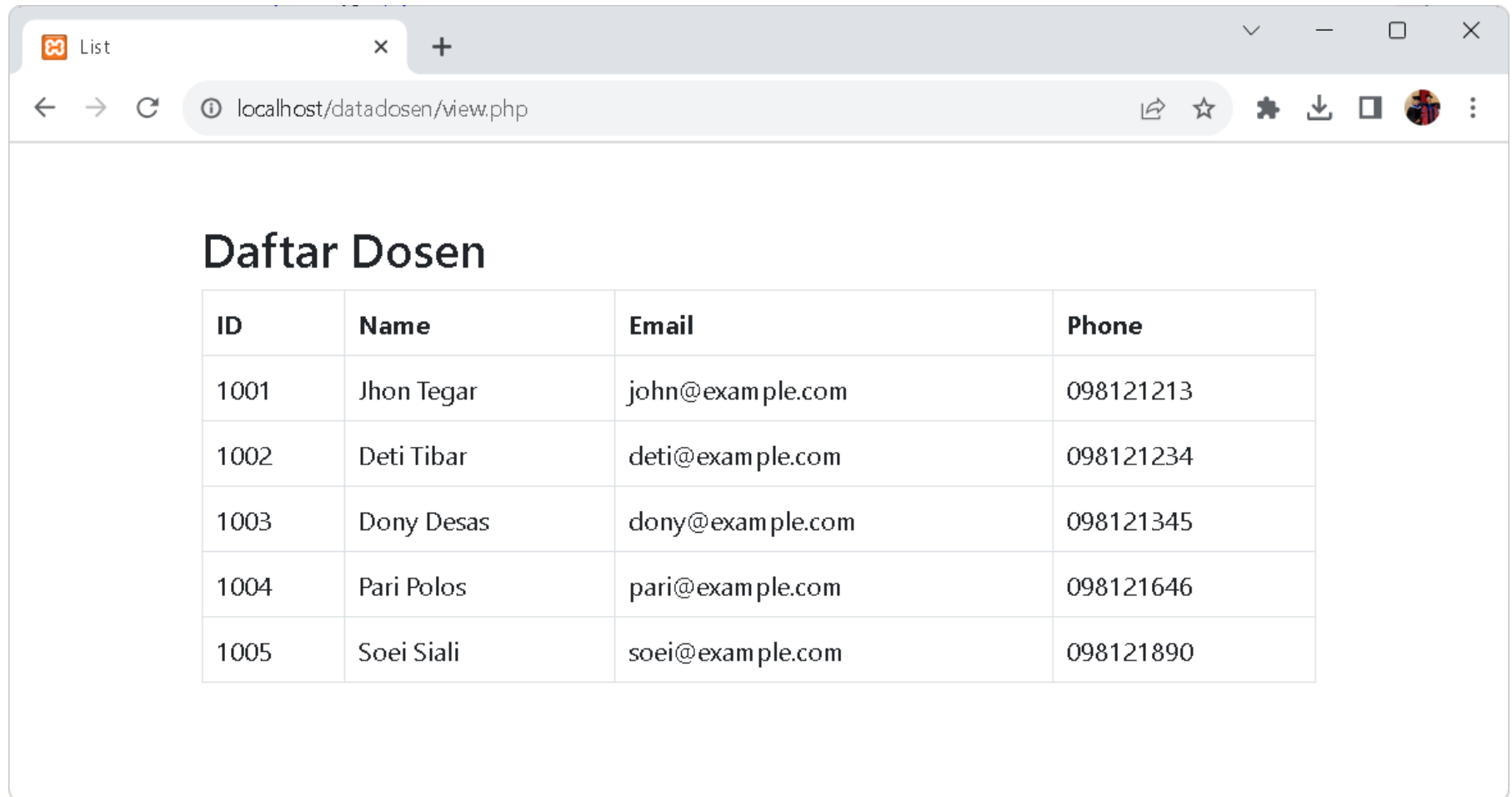
```
?>
```

```
</tbody>
```

Expected Output

Open browser dan ketikan URL berikut:

<http://localhost/datadosen/view.php>



The screenshot shows a web browser window with a single tab titled 'List'. The address bar displays 'localhost/datadosen/view.php'. The page content features a heading 'Daftar Dosen' followed by a table with four columns: ID, Name, Email, and Phone. The table contains five rows of data for different lecturers.

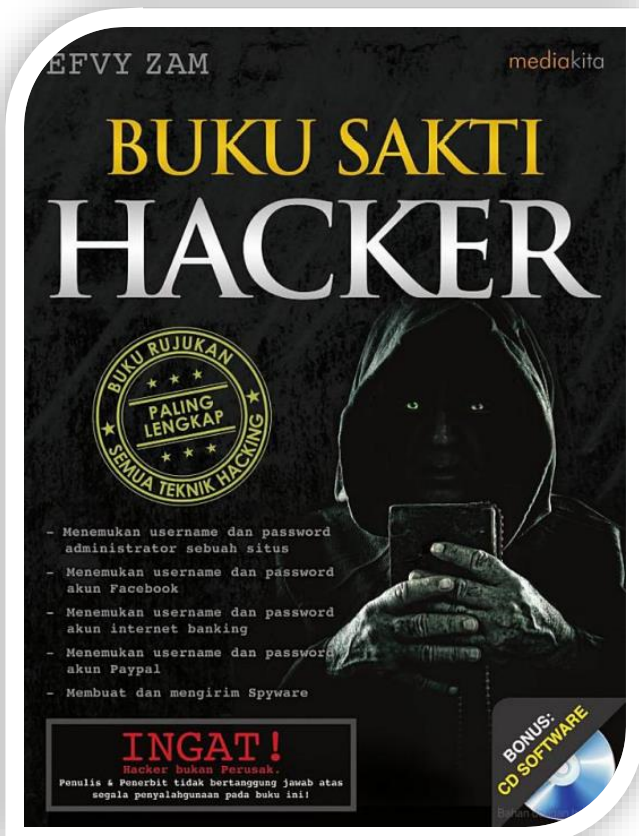
ID	Name	Email	Phone
1001	Jhon Tegar	john@example.com	098121213
1002	Deti Tibar	deti@example.com	098121234
1003	Dony Desas	dony@example.com	098121345
1004	Pari Polos	pari@example.com	098121646
1005	Soei Siali	soei@example.com	098121890

5 – Analyze access.log, error.log and php_error_log Files

Apakah ada sistem yang benar-benar aman 100%?

Port Scanning

4



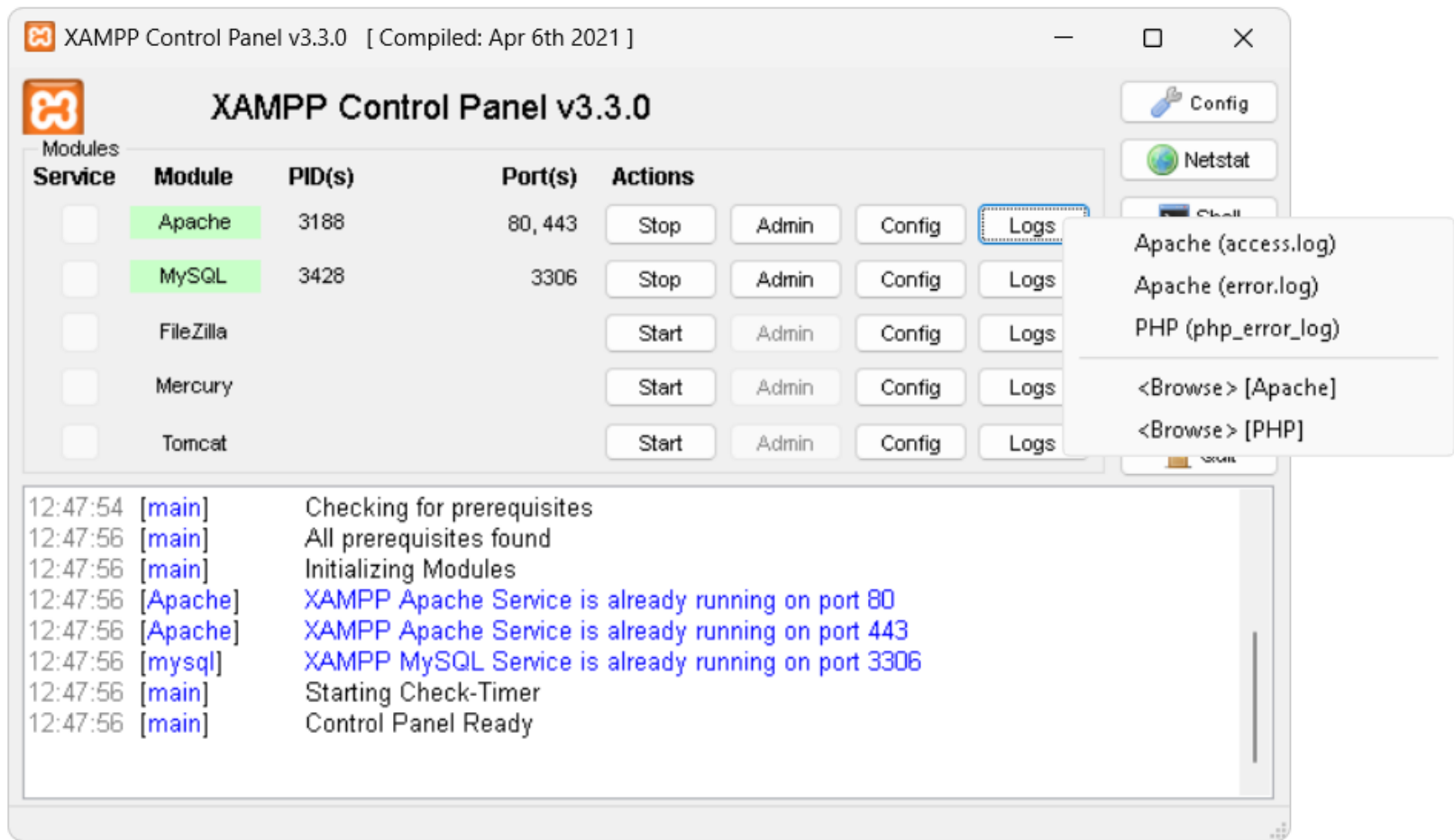
Jika diibaratkan sebuah rumah, port adalah pintu dan jendela rumah tempat keluar masuknya data. Secara logika, tidak ada sistem yang aman 100%. Apabila sebuah sistem aman 100%, tentu saja semua pintu dan jendela akan ditutup semua. Ibaratnya, jika pintu internet tidak dibuka, Anda pun tidak akan bisa menghubungkan komputer dengan internet. Jadi, bisa kita katakan bahwa port adalah pintu keluar masuknya paket data. }

Secara garis besar, port dapat dibagi dua bagian. Yang pertama adalah port fisik (*physical port*) yang merupakan port di bagian belakang CPU, seperti port serial, dan port monitor. Lalu ada pula port perangkat lunak (*software port*), merupakan port yang digunakan oleh software untuk melakukan koneksi dengan komputer lain.

Port juga mengidentifikasi sebuah proses tertentu dimana sebuah server dapat memberikan sebuah layanan kepada klien atau bagaimana sebuah klien dapat mengakses sebuah layanan yang ada dalam server. Ada banyak port yang terdapat pada sebuah komputer, apalagi sewaktu terhubung dengan internet. Beberapa port yang umum adalah port 80 (HTTP), yaitu port untuk membuka sebuah halaman website, port 20 (FTP) untuk melakukan upload maupun download file, port 110 (POP3) untuk menerima email. Serta masih banyak jenis port lainnya.

```
semmy@presensi: ~  
Sep 16 00:07:51 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:54 presensi phpMyAdmin[18162]: message repeated 4 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:07:55 presensi phpMyAdmin[18162]: user denied: phpmyadmin (mysql-denied) from 192.3.228.147  
Sep 16 00:07:55 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:56 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:56 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:07:57 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:57 presensi phpMyAdmin[18162]: user denied: web (mysql-denied) from 192.3.228.147  
Sep 16 00:07:58 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:59 presensi phpMyAdmin[18162]: message repeated 3 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:00 presensi phpMyAdmin[18162]: user denied: wordpress (mysql-denied) from 192.3.228.147  
Sep 16 00:08:00 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:02 presensi phpMyAdmin[18162]: message repeated 3 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:02 presensi phpMyAdmin[18162]: user denied: developer (mysql-denied) from 192.3.228.147  
Sep 16 00:08:03 presensi phpMyAdmin[18162]: user denied: dbs (mysql-denied) from 192.3.228.147  
Sep 16 00:08:03 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:04 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:04 presensi phpMyAdmin[18162]: user denied: wp (mysql-denied) from 192.3.228.147  
Sep 16 00:08:05 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:09 presensi phpMyAdmin[18162]: message repeated 8 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:09 presensi phpMyAdmin[18162]: user denied: newuser (mysql-denied) from 192.3.228.147  
Sep 16 00:08:10 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:10 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:11 presensi phpMyAdmin[18162]: user denied: database (mysql-denied) from 192.3.228.147  
Sep 16 00:08:11 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:12 presensi phpMyAdmin[18162]: message repeated 2 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:13 presensi phpMyAdmin[18162]: user denied: user (mysql-denied) from 192.3.228.147  
Sep 16 00:08:14 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:15 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:16 presensi phpMyAdmin[18162]: user denied: newuser (mysql-denied) from 192.3.228.147  
Sep 16 00:08:16 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:17 presensi phpMyAdmin[18162]: user denied: dev (mysql-denied) from 192.3.228.147  
Sep 16 00:08:17 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:18 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:19 presensi phpMyAdmin[18166]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:20 presensi phpMyAdmin[18166]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:20 presensi phpMyAdmin[18166]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:21 presensi phpMyAdmin[18166]: user denied: phpmyadmin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:22 presensi phpMyAdmin[18166]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:26 presensi phpMyAdmin[18166]: message repeated 7 times: [ user denied: root (mysql-denied) from 192.3.228.147]
```

Hardening is an *approach in computer security* that aims to reduce security risks by reducing potential system vulnerabilities. While no solution can guarantee complete security, hardening is still an important step to protect systems and data.



Open and analyze **error.log**, **access.log** and **PHP (php_error_log)** files to find server activities and provide important information related to Apache operations. Students must be able to analyze all error messages or warnings, user access patterns on the server generated by the Apache server, monitoring server activity to detect potential abuse (*brute-force attacks or unauthorized access*) and provide important information as a report in the class.

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

